ARPES viewer — User Guide

中文版 / Chinese version · *Version française*

Program version 2026-10-12 · source code: https://github.com/DingPei1995/ARPES-data-browser

This guide has four parts:

- **Part I — Installation and setup**: the version, where the code is, what data the program reads, and how to set up Python and start it.
- **Part II — Looking at data during the experiment**: the main window, the right-click menus, and the viewers for SPEM, map and cut data. These are the tools for deciding quickly, at the beamline, where to measure next.
- **Part III — Advanced processing for publication**: de-gridding, the Fermi level, k conversion, the Brillouin zone, kz, curvature, dichroism, peak fits and figures, in the order a paper's data usually goes through them.
- **Part IV — Troubleshooting**: autosave and crash recovery, memory, and answers to common problems.

Every screenshot was taken from the running program on the synthetic data written by `devtools/demo_data.py`, so you can follow along without a beamtime (§3). The formulas, tests and design choices behind each step are in the README and the CHANGELOG.

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Part I — Installation and setup

1. About the program

Version	2026-10-12 (last modified 12 October 2026). The version is also at the top of the README; CHANGELOG.md lists every change.
Source code	https://github.com/DingPei1995/ARPES-data-browser (the program is in the `ARPES_viewer/` folder)
Written in	Python 3, PyQt5 and pyqtgraph; numpy, scipy and h5py for the numerics and the files
Platforms	Windows, macOS and Linux — anywhere PyQt5 runs

Data the program reads

Beamline / source	Files	Opens as
SOLEIL **ANTARES**	NeXus/HDF5 `*.nxs`	cuts, deflector maps, real-space (SPEM) scans
SOLEIL **CASSIOPEE**	Scienta SES text files: single spectra, and numbered folders	cuts; a folder becomes one map or one photon-energy (kz) scan
CASSIOPEE **spin end station** (MBS A-1)	`*.krx` and `*.txt`	cuts, deflector maps, spin-resolved EDCs
This program	its own saved `*.nxs`	everything it saves, with the steps that made it

Adding a beamline means adding one reader file to `loader/` (see *Adding a beamline* in the README).

Data kinds

Every dataset in the program has one of these kinds. The kind decides which viewer opens it and which functions it is offered.

Kind	Listed as	Axes	What it is
`cut`	`[Cut]`	angle (or k) × E	one spectrum
`map`	`[Map]`	angle × angle × E	a deflector or polar-angle scan
`k_map`	`[k-map]`	kx × ky × E	a map converted to momentum
`kz_map`	`[kz map]`	hν × angle × E	a photon-energy scan
`kz_map_k`	`[kz map (k)]`	k_z × k∥ × E	that scan converted to momentum
`spem_4d`, `spem_1d`	`[SPEM]`	x (× y) × angle × E	real-space (nano-ARPES) scans
`edc`, `mdc`, `spin_edc`	`[EDC]`, `[MDC]`, `[Spin EDC]`	E (or k) × channels	curves, with optional σ channels

2. Installing Python and starting the program

Python. Use Python **3.9 or newer** (it has been tested up to 3.13). Any distribution works: python.org, Anaconda/Miniconda, or the system Python on Linux.

Packages. Five are needed:

```
pip install PyQt5 pyqtgraph numpy h5py scipy
```

It is cleanest to install them into an environment of their own, so the program does not depend on, or upset, other Python work on the machine:

```
# with venv
python -m venv arpes-env
source arpes-env/bin/activate          # Windows: arpes-env\Scripts\activate
pip install PyQt5 pyqtgraph numpy h5py scipy

# or with conda
conda create -n arpes python=3.12 pyqt pyqtgraph numpy h5py scipy -c conda-forge
conda activate arpes
```

Versions it was last tested with: PyQt5 5.15, pyqtgraph 0.14 (0.13 works too), numpy 2.3, h5py 3.16, scipy 1.18.

Getting the code. Download it from the GitHub page (*Code → Download ZIP*), or:

```
git clone https://github.com/DingPei1995/ARPES-data-browser.git
cd ARPES-data-browser/ARPES_viewer
```

Starting. From the `ARPES_viewer` folder:

```
python ARPES_viewer.py
```

Settings and working files are kept in the folder `.arpes_viewer` in your home directory:

File	What it holds
`~/.arpes_viewer/config.json`	the last data folder, and the colormap a new window starts with
`~/.arpes_viewer/sessions/`	the autosaved working copies of computed data (§23)
`~/.arpes_viewer/sessions/operations.log`	one line for every autosave, save and deletion

Deleting `config.json` resets the settings; nothing else is stored.

Checking the installation. The test suite runs without a display:

```
python -m pytest           # from the ARPES_viewer folder; needs pip install pytest
```

3. Trying it with demo data

```
python devtools/demo_data.py           # writes ./demo_data/*.nxs
```

This writes six files of synthetic data: a Fermi-surface map, a pair of cuts taken with LH and LV polarisation, a photon-energy scan, a spin-resolved EDC, a small real-space scan of a flake with two domains, and a map with a detector mesh printed on it. They come from a toy band structure (a hexagonal electron pocket and a hole band at Γ, hole pockets at K) with a Fermi cut-off, the analyser resolution and Poisson noise, and open exactly like measured data. Point the main window's folder box at `demo_data/` and add them.

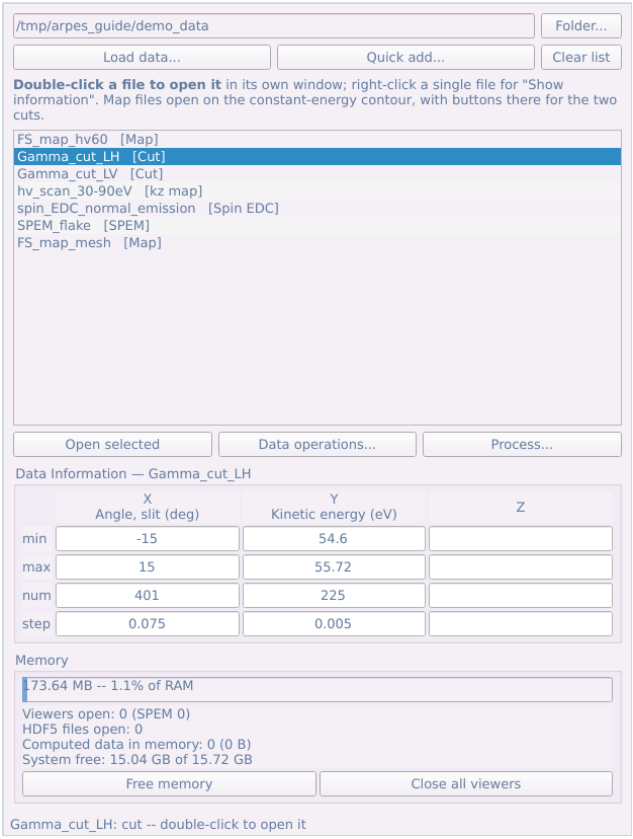
To re-take every screenshot in this guide after changing a window:

```
python devtools/guide_screenshots.py    # writes docs/images/*.png
python devtools/build_guide_pdfs.py     # rebuilds the PDFs (needs pandoc and playwright)
```

Part II — Looking at data during the experiment

4. The main window and its right-click menu

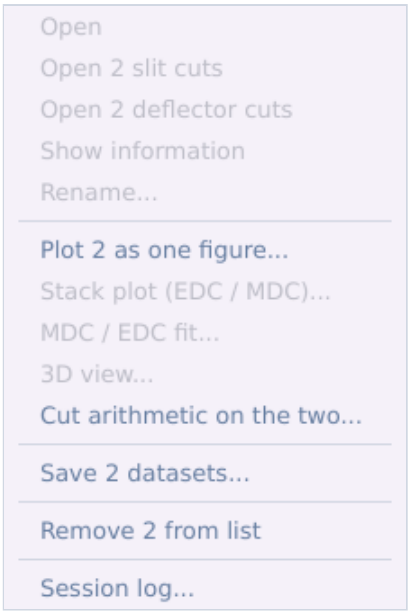
The main window is a **browser**: one row per *dataset*, not per file. One `.nxs` file often holds several measurements, and each gets its own row. The kind is written after the name: `[Cut]`, `[Map]`, `[kz map]`, `[k-map]`, `[SPEM]`, `[EDC]`, `[Spin EDC]` …



To …	do this
open a dataset in its own window	**double-click** the row, or **Open selected**
work on one or several datasets	select them and **right-click** (menu below)
read the axes: first and last value, points, step	right-click → **Show information** (fills *Data Information*)
truncate, normalise or bin several datasets at once	**Data operations...** (§18)
smooth, differentiate, curvature, background, symmetrise	**Process...** (§18)
see and free memory	the **Memory** box at the bottom (§23)

The list's right-click menu

Select one or more rows and right-click. The menu acts on the **whole selection** (shift-click and ctrl-click select several rows):



Entry	Works on	What it does
Open	one dataset	opens it in its viewer — the same as a double-click
Open slit cut / **Open deflector cut**	maps (one or several)	opens just that cut of each map, without the contour
Show information	one dataset	kind, axes and full metadata in the information window
Rename...	one dataset	changes the name it is listed and saved under (the file on disk keeps its name)
Plot as a figure...	cuts or curves	one panel per dataset in the figure composer (§22)
Stack plot (EDC / MDC)...	one cut	a waterfall of EDCs or MDCs (§9)
MDC / EDC fit...	one cut in k	the peak-fit panel (§20)
3D view...	one map	slices, notched cube and projections (§11)
Cut arithmetic on the two...	exactly two cuts	dichroism, ratios, differences (§19); the first selected is A
Save dataset(s)...	any	writes the selection to one `.nxs` file of your own
Remove from list	any	takes the rows out of the list, after asking; files on disk are never deleted
Session log...	—	what has been autosaved, saved and deleted, and where (§23)

The menu is decided from the selection alone, without reading any data, so it opens at once however large the files are. An entry that does not fit the selection is **greyed out**; hover over it and its tooltip says why (for example, *Cut arithmetic* needs exactly two cuts, the *3D view* needs a map).

Show information also opens the full metadata of the dataset in a window of its own, with a filter box and **Export metadata (.csv)**:



Clear list asks first, and offers to save anything computed in this session that has not been saved yet. Files on disk are never deleted.

5. Loading data

Load data... opens the loader. **Quick add...** skips it and adds the chosen files with everything detected automatically — the fast way during a beamtime.

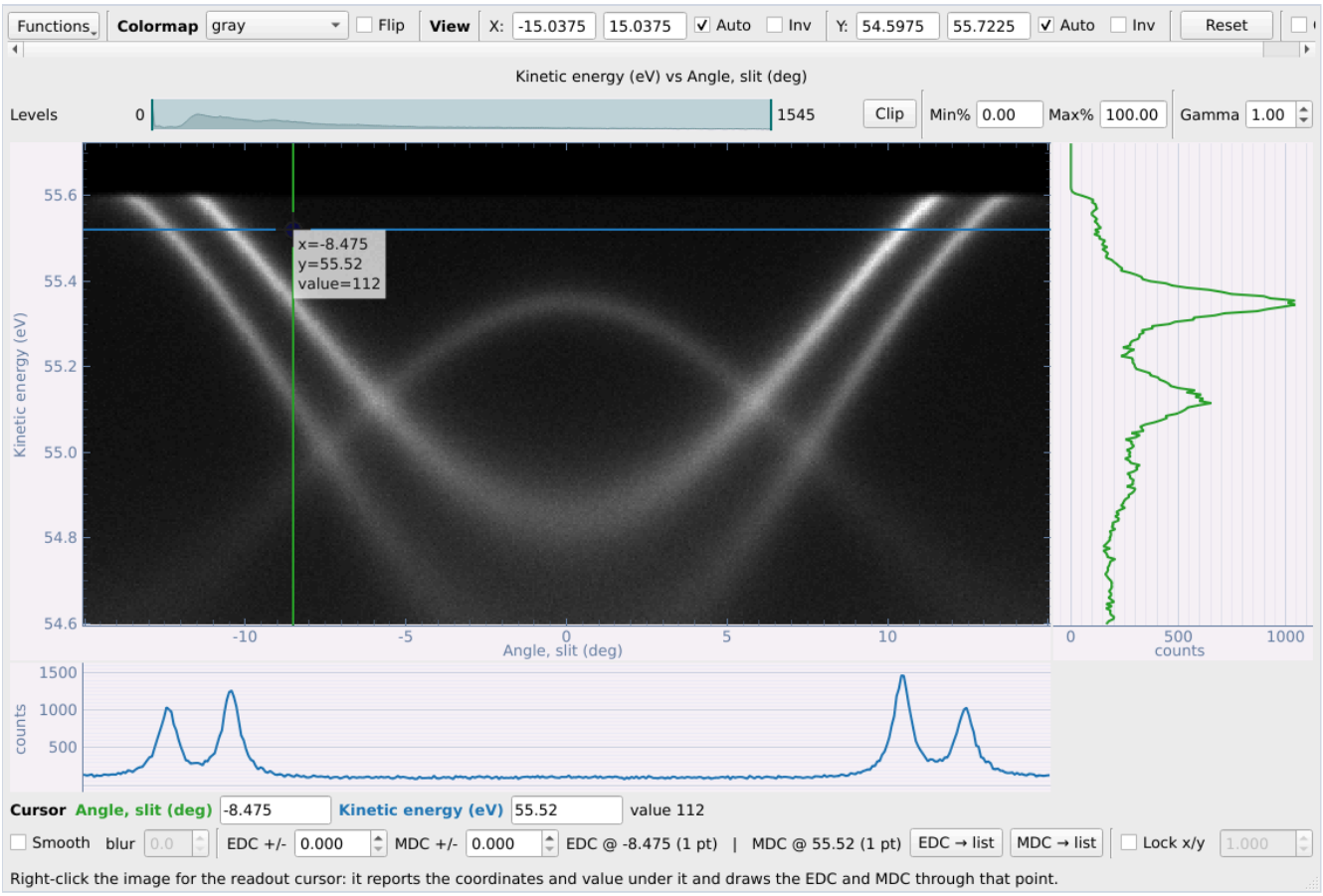


1. **Files** — pick one or many (*Select file(s)...*). The file filter is built from what the installed readers can open.
2. **Reader** — detected automatically, and the result is shown (*Detected: 3 × This program's own format*). Override it only if a file is being misread.
3. **Datasets found** — what is inside each file, read from its *structure* only. This is instant even for a multi-GB scan; nothing is read until you open a row.
4. **Axes** — *Axis order* reorders the array's dimensions once, at the door. *First axis of a map is* says what was scanned: a deflector angle (the default), a **photon energy**, a temperature, a gate voltage, a delay... This keeps the right unit everywhere afterwards, and stops a k conversion being offered for an axis that is not an angle.

CASSIOPEE folders. Select any file of a numbered series (`name_1_ROI1_.txt`, `name_2_ROI1_.txt`, ...) and the whole folder is listed once, as one assembled cube. Whether it becomes an angle map or a kz map is read from the `_i` parameter files next to the spectra.

6. Image windows: controls and the right-click menu

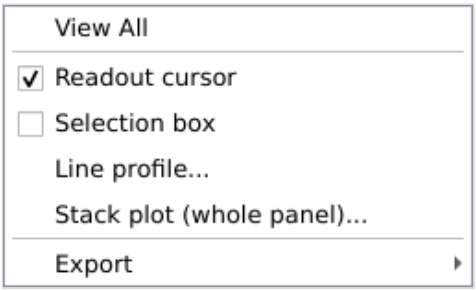
Each dataset opens in its own window, so a map, its cuts and another file can sit side by side.



- **Top row**: the **Functions** menu (everything the window can do), the colormap (per window) with **Flip**, and the view controls — axis ranges, **Auto**, **Inv** (invert), **Reset**, **Grid**.
- **Above each image**: the **Levels** bar (drag the two handles, or type **Min%/Max%** percentiles and press **Clip**; *right-click* **Clip** for ready-made pairs: 0.5–99.5 %, 1–99 %, 2–98 %, 5–95 %, 0–100 %) and **Gamma**.
- **Under each image**: **Smooth** (display only; the data are not changed) and **Lock x/y** (aspect ratio).
- **Mouse**: drag to pan, wheel to zoom, **right-drag** to zoom each axis separately.

The image right-click menu

Right-click on any image:



Entry	What it does
View All	fits the view back to the whole image
Readout cursor	a crosshair you drag. The **Cursor** row under the image shows its position and the value there; type a position and press Enter to put it exactly there. On a cut it also draws the **EDC** (right) and **MDC** (below) through that point; **EDC ±** / **MDC ±** integrate over a window, shown as a shaded band.

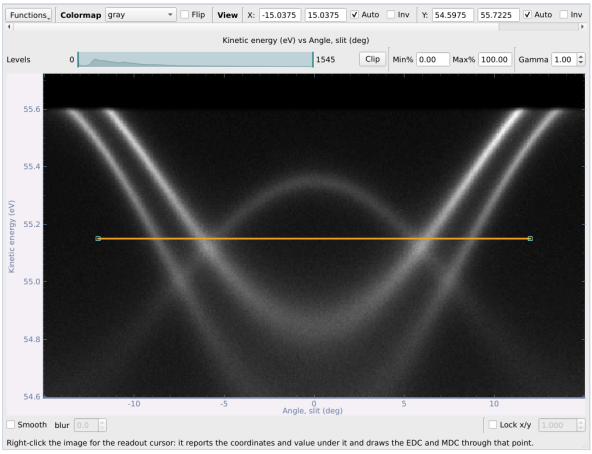
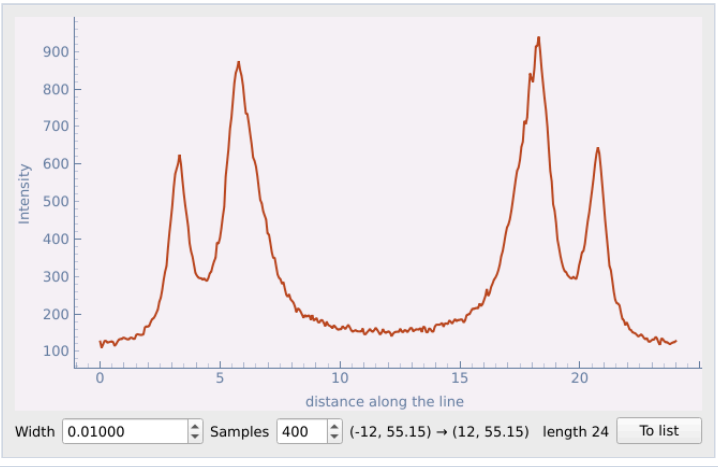
Entry	What it does
Selection box	a rectangle with two corner handles. Its corners appear in the **Selection** row under the image, where they can be typed (then **Refresh**). In a SPEM window the box is what you integrate over (§7); everywhere it sets the range of the stack plot.
Line profile...	the intensity along a line at any angle, with an integration width (below)
Stack plot of the selection...	a waterfall of EDCs or MDCs across the selection box — or across the whole panel when there is no box. The entry shows the range it will use.
Export → Data (.nxs) / Image / Axes only (.eps)	saves exactly what this panel shows (§22)

SPEM windows add three entries of their own, described in §7. pyqtgraph's own menu items (mouse mode, axis submenus, plot options, its export dialog) are removed on purpose: what was useful in them is on the visible view bar.

EDC → list / **MDC → list** (next to the curves) put the curve on screen into the main list as a dataset of its own (§10). **Functions → Slice → Save slice to the main list** / **Open slice in a new panel** keeps a copy of the slice on screen.

Line profile

Right-click → Line profile... puts an orange line across the image and opens a window with the intensity along it. Drag the line, or its ends; **Width** integrates that far either side (in the units of the vertical axis), **Samples** sets the number of points, and **To list** keeps the profile as a dataset. Closing the window removes the line. Use it for a band that runs at an angle, where neither the EDC nor the MDC follows it.

The line on the cut	The profile along it
	

The Functions menu

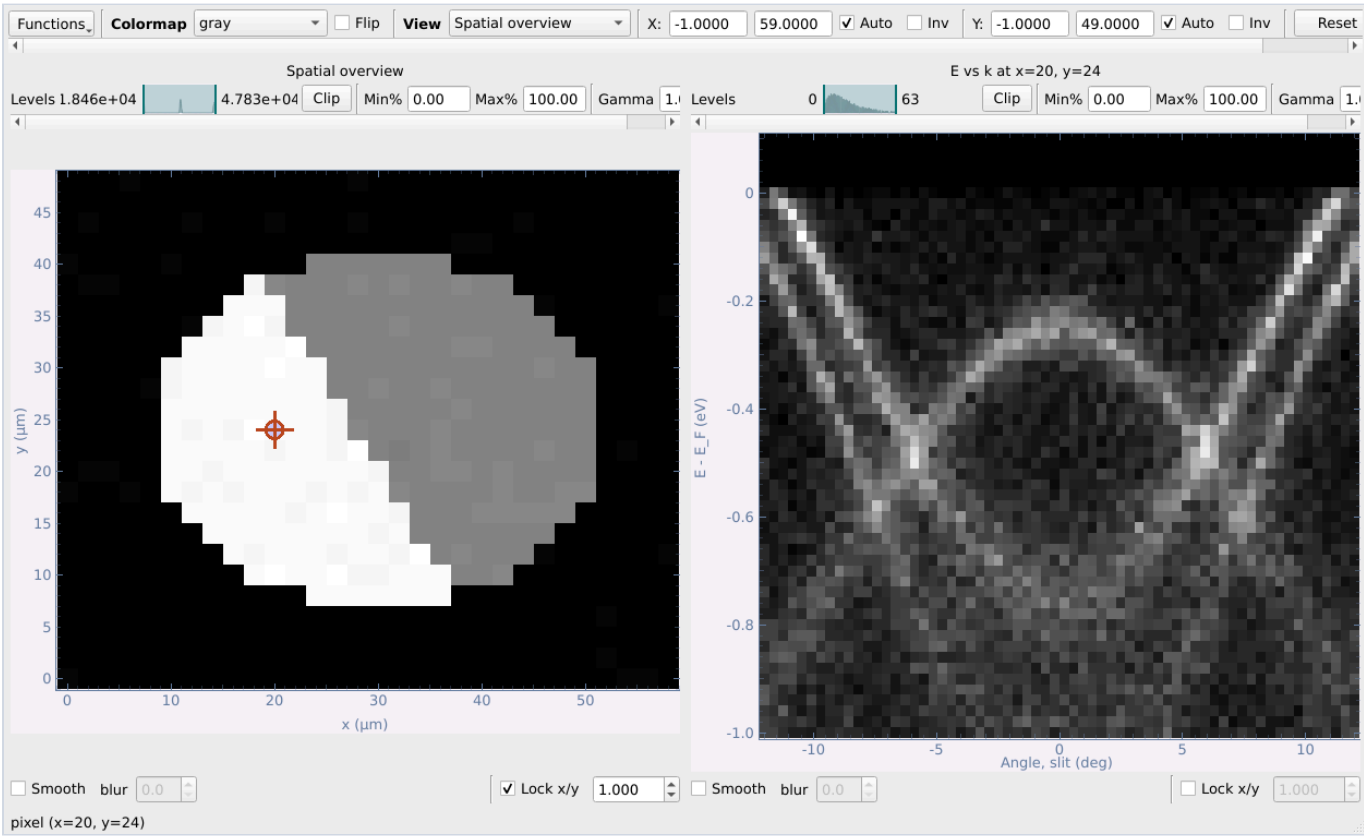
Viewer	Analysis	Data operations	Visualization
map / kz map / k-map		Arbitrary cut, Map k conversion, kz map processing, kz → momentum, De-grid map	Brillouin zone (k-maps), Slice figure
cut	Fermi level, MDC / EDC fit	FS correction, Cut k conversion, Cut arithmetic, De-grid	

Viewer	Analysis	Data operations	Visualization
curve	Curve fit, Spin analysis	Crop, Bin, Normalise, Background, Shift axis, Add √N errors	As a figure

An entry that does not apply is not shown; one that applies but cannot run yet is greyed, with the reason in its tooltip.

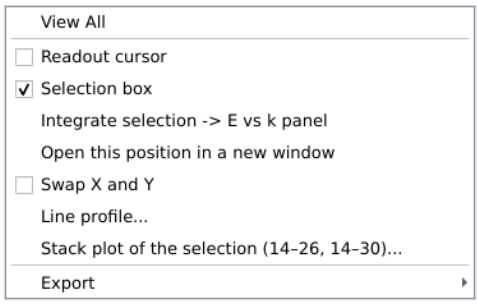
7. SPEM (real-space) scans: finding the spot to measure

A real-space scan (nano-ARPES, `[SPEM]`) opens two panels side by side: the **spatial map** on the left — the total intensity at each sample position — and the **E vs k spectrum** on the right. The red target on the map chooses the pixel whose spectrum is shown, and the right panel's title says which pixel it is (*E vs k at x=20, y=24*).



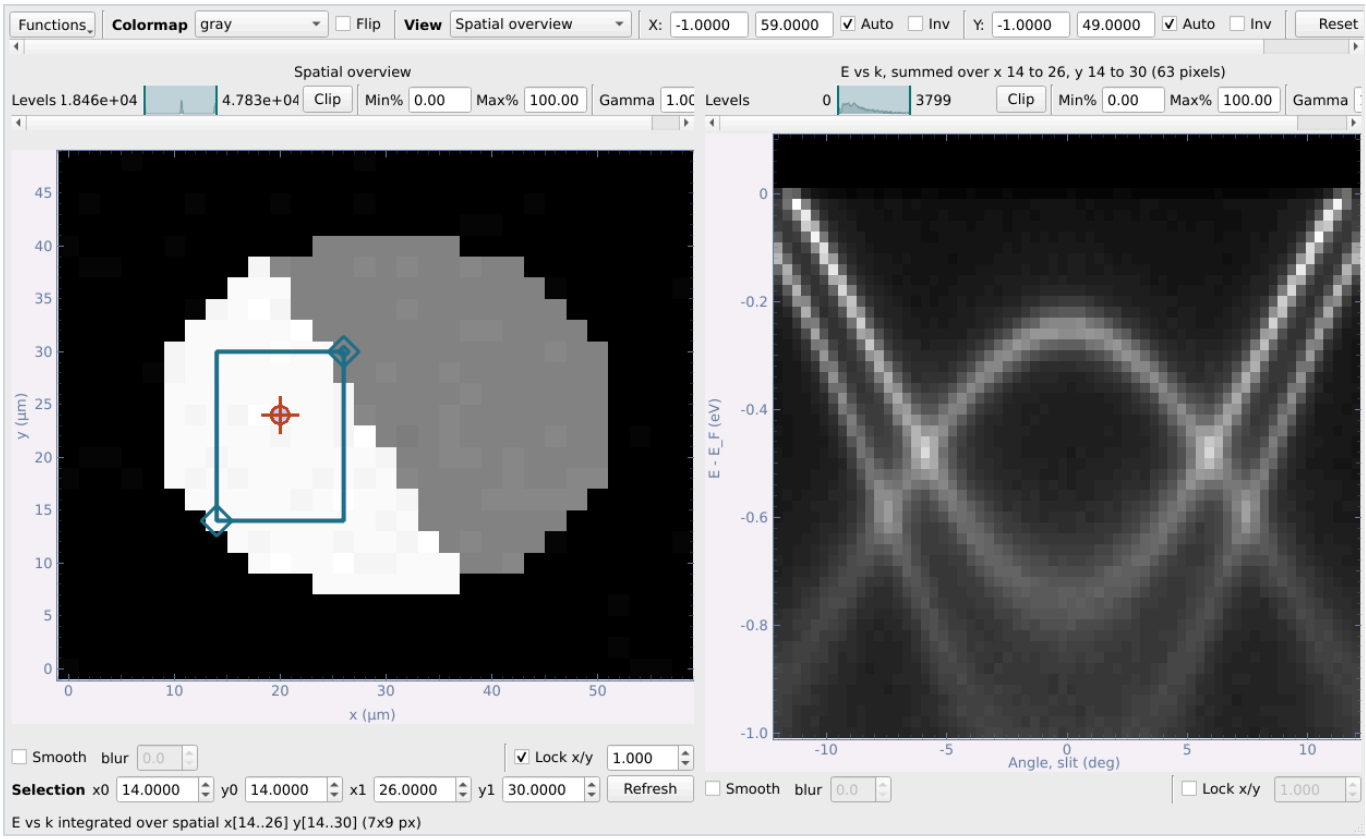
Drag the red target to read the spectrum anywhere on the sample. This is the quickest way to see whether a flake shows bands at all, and how sharp they are.

Right-click on the spatial map

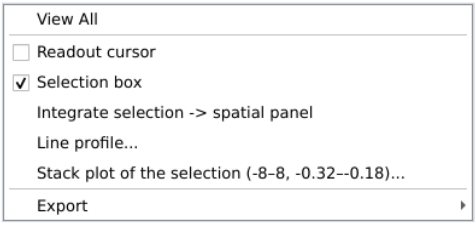


Entry	What it does
Selection box	draws a box on the map
Integrate selection -> E vs k panel	the spectrum **summed over every pixel in the box**: more counts, and the average over one region of the sample
Spatial map over the whole spectrum	appears after a spectrum box has been integrated onto the map (below): goes back to the map summed over all k and E
Open this position in a new window	a snapshot of the spectrum shown now, in a window of its own — keep several open to compare spots
Swap X and Y	draws the map transposed, for a lab whose convention has the other stage horizontal. The cursor and the selection box stay on the same sample pixels; only the drawing changes.
Readout cursor, Line profile, Stack plot, Export	as on every image (§6)

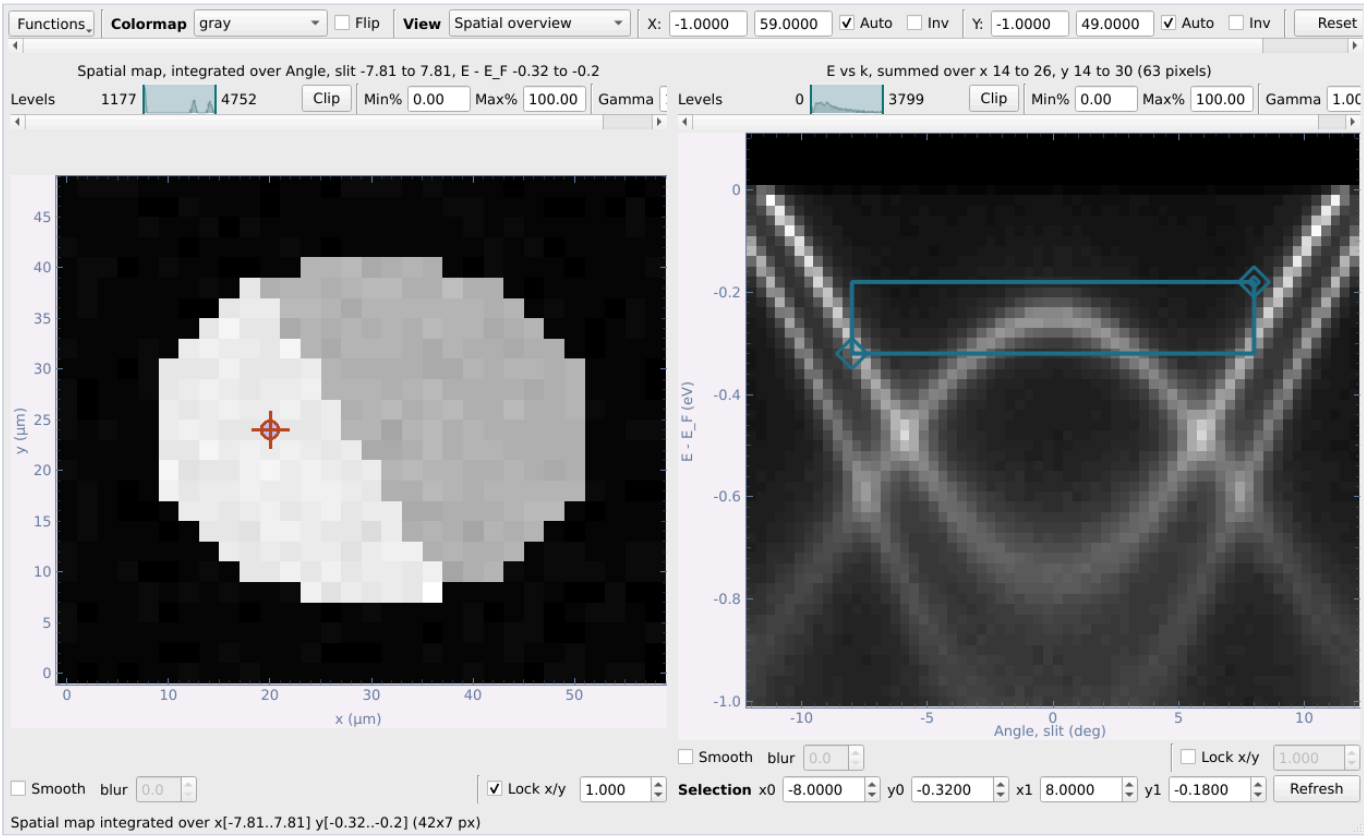
Integrating a region of the sample. Right-click the map → **Selection box**, drag the box over the region (or type its corners in the **Selection** row and press **Refresh**), then right-click → **Integrate selection -> E vs k panel**. The spectrum is now the sum over the box, and its title says so (*E vs k, summed over x 14 to 26, y 14 to 30 (63 pixels)*):



Right-click on the spectrum: where on the sample does a feature come from?



The opposite direction. Right-click the spectrum → **Selection box**, frame a feature in (k, E) — a band, a shoulder, the Fermi edge of one phase — and right-click → **Integrate selection -> spatial panel**. The map is redrawn as the intensity of **that window only** at every position, and its title says which window. Here the box frames a band that only one domain of the flake has, and the map lights up exactly that domain:



This is how domains, flake edges, terraces or a second phase are found. To get the full map back, right-click the map → **Spatial map over the whole spectrum**.

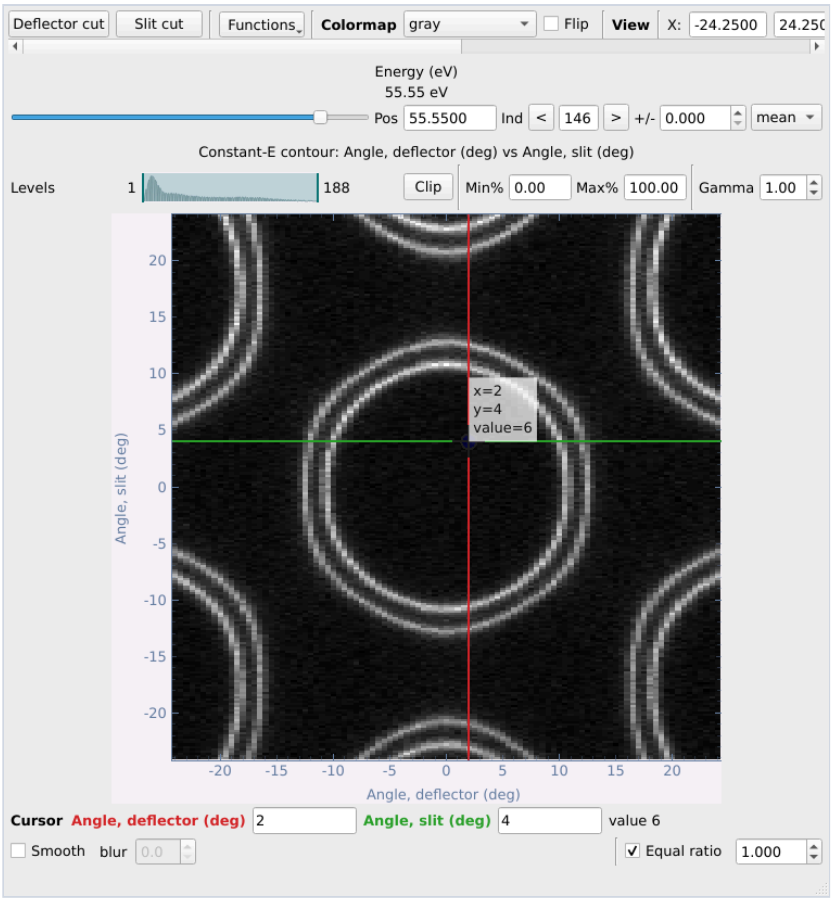
A practical routine at the beamline:

1. Open the overview scan; drag the red target over the flake and look at the spectra.
2. Box a feature in the spectrum → integrate onto the map to see where that feature is strongest.
3. Box the best region of the map → integrate onto the spectrum to check the bands with good statistics.
4. **Open this position in a new window** for the candidates, compare them side by side, and read the coordinates off the cursor row for the next measurement.

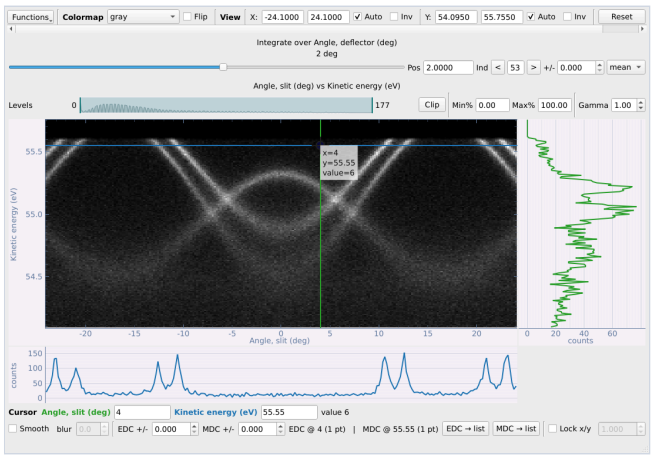
A line scan (`spem_1d`) opens the same way, with position × angle on the left; the box integration needs a 2-D map, so it is offered only for 2-D scans.

8. Maps: the contour and its two cuts

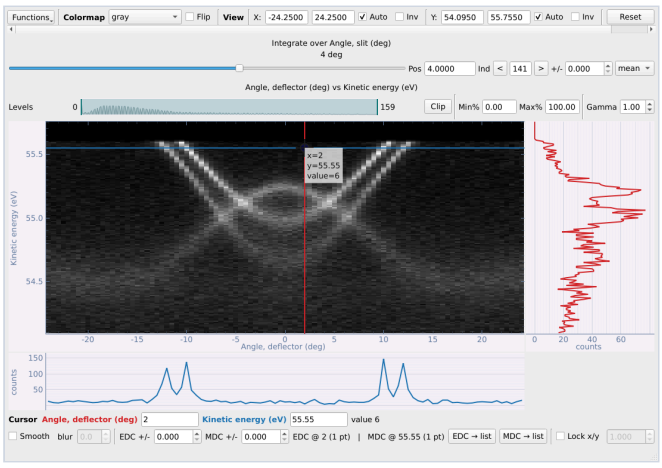
A map opens on its **constant-energy contour**. The slider at the top picks the energy (**Pos** in eV, or **Ind** as an index; **±** integrates over a window, **mean** or **sum**). **Deflector cut** and **Slit cut** open the two orthogonal cuts, each in its own window.



Slit cut (slit angle vs energy)



Deflector cut (deflector angle vs energy)



One cursor for all three windows. Switching the readout cursor on in any of them (right-click → *Readout cursor*) switches it on in the others, and moving it anywhere re-slices the others through the new point. The axes keep one colour each everywhere: **deflector red, slit green, energy blue**. A cursor line has the colour of the axis it holds constant, and the EDC or MDC it feeds is drawn in the same colour. Integration widths (the ± boxes) belong to each window and are never shared.

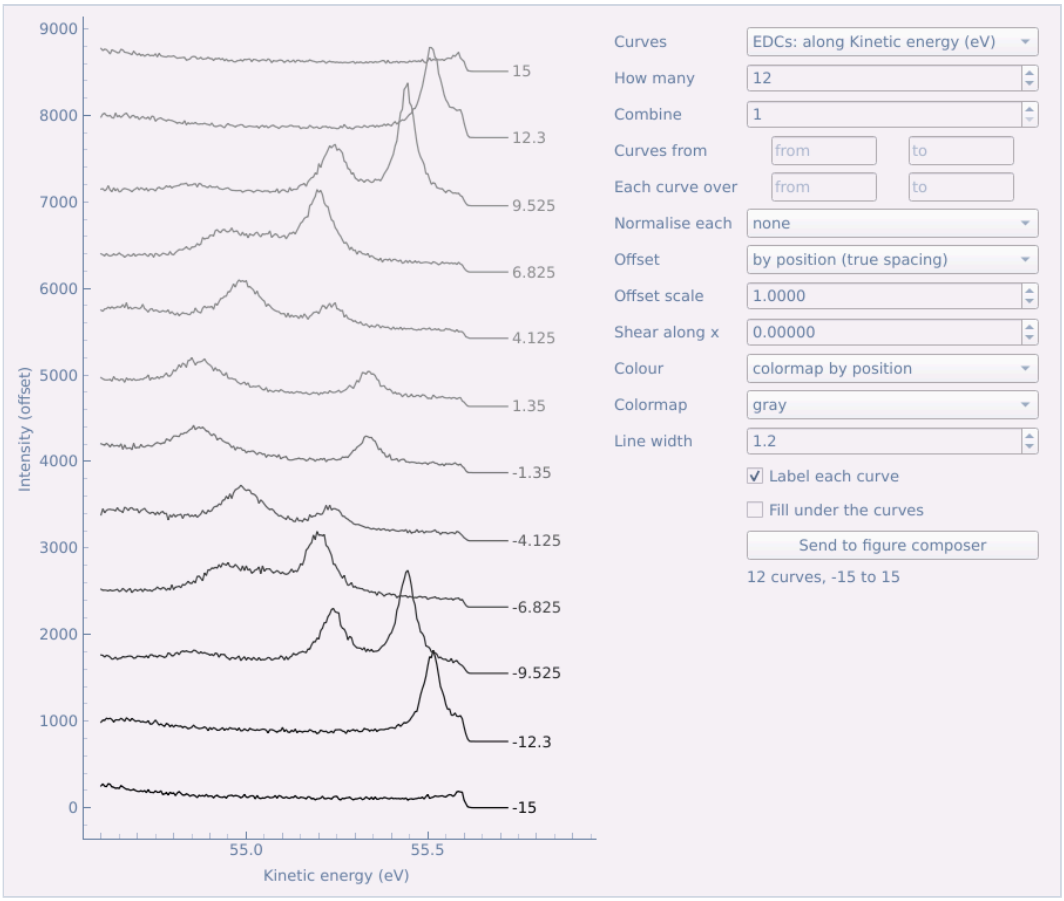
From the list you can also open a map's cut straight away (right-click → **Open slit cut** / **Open deflector cut**); the contour then stays hidden until you ask for it (*Functions* → *Show the map*).

During a beamtime this answers *is the sample aligned, where is Γ, which direction is the cut going to follow* — before any conversion.

9. Cuts: quick looks

A cut opens in a single image panel (§6). The right-click tools give fast answers:

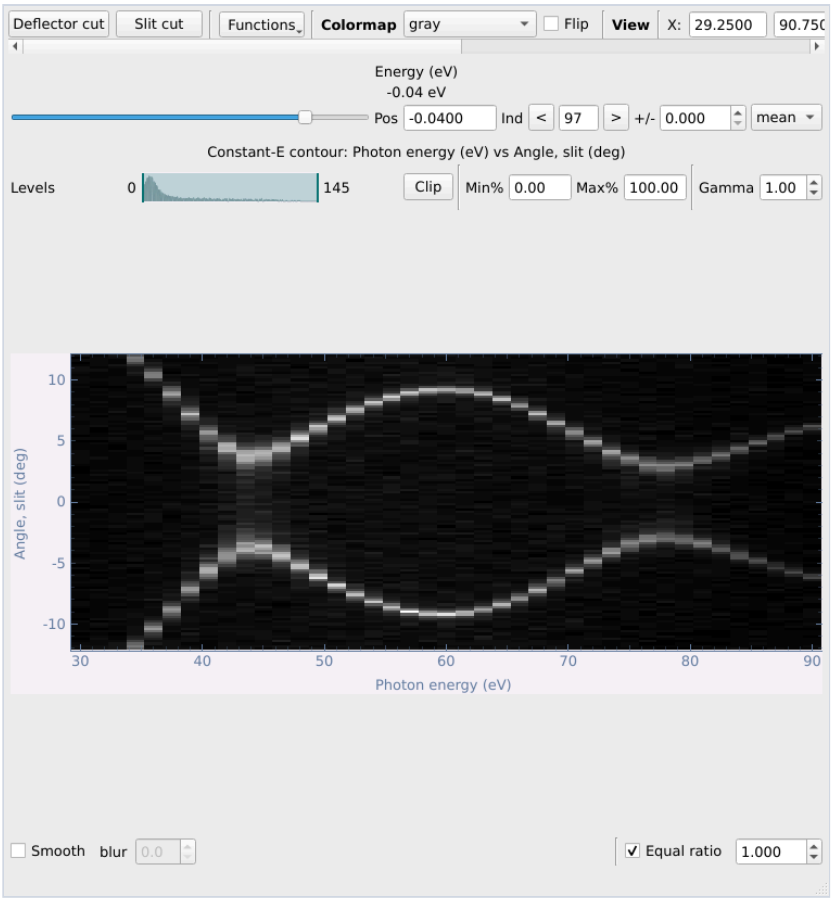
- **Readout cursor** with **EDC ±** / **MDC ±**: the curves through any point, integrated over a window.
- **Line profile...**: the intensity along any line.
- **Stack plot**: right-click the cut (or its row in the list) → **Stack plot of the selection...**. The curves run across the selection box, if there is one: a box taller than it is wide gives EDCs across its angle range, a wide one MDCs across its energy range. The window sets the number of curves, binning, normalisation, offset, shear and colours; **Send to figure composer** makes it a figure panel.



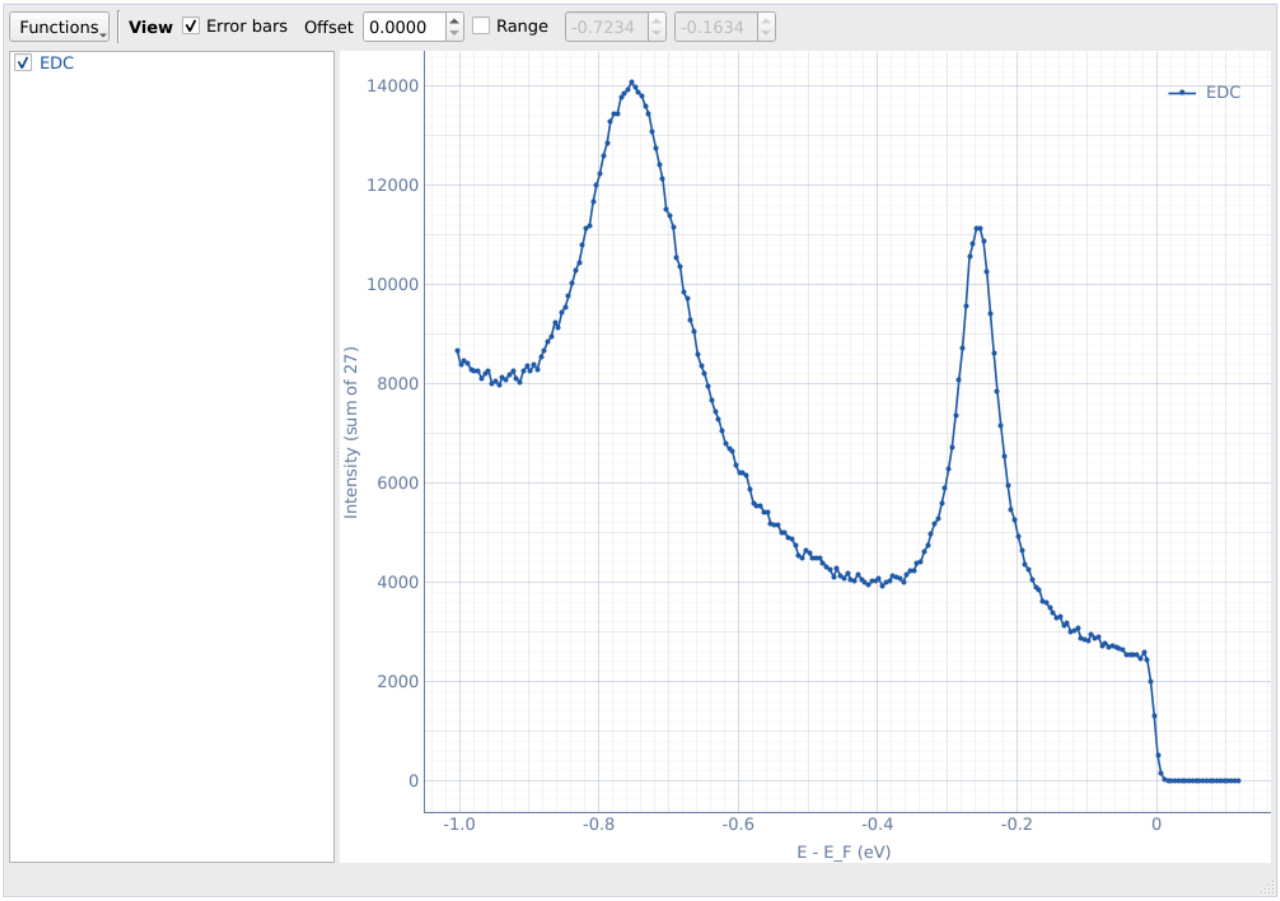
- **Fermi level**: *Functions → Fermi level...* gives E_F and the resolution in a few seconds (§14) — useful to check the energy calibration right after a photon-energy change.

10. Photon-energy scans, curves and spin EDCs: quick looks

Photon-energy scans. A photon-energy scan is listed as a `kz map`; its contour is photon energy against angle at one energy, and its cuts open like a map's. Before processing (§17) the Fermi level of each spectrum is not aligned, so treat it as a quick look.

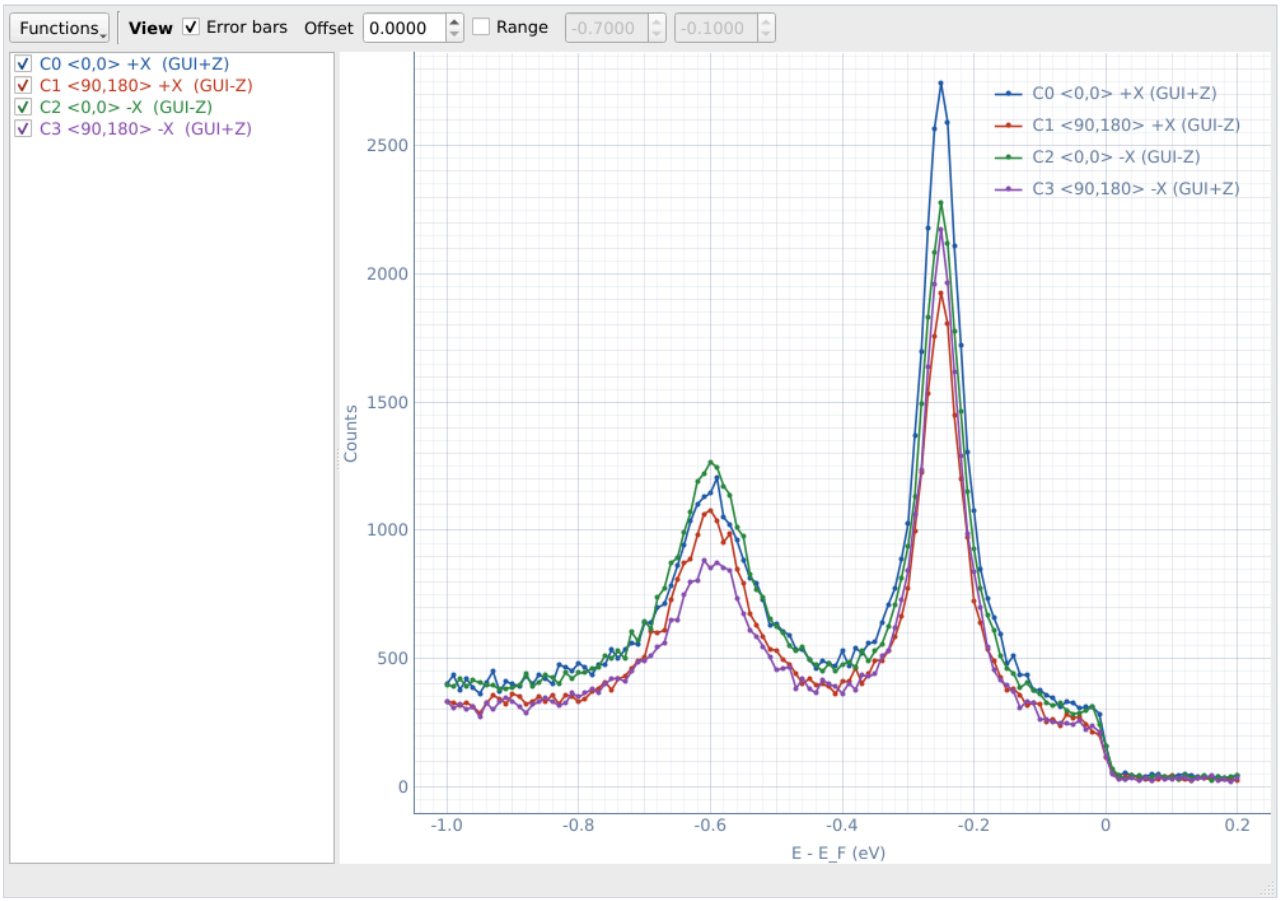


Curves. EDC → list / **MDC → list** under any image, or a spin file, give a curve dataset, which opens in the **curve viewer**:



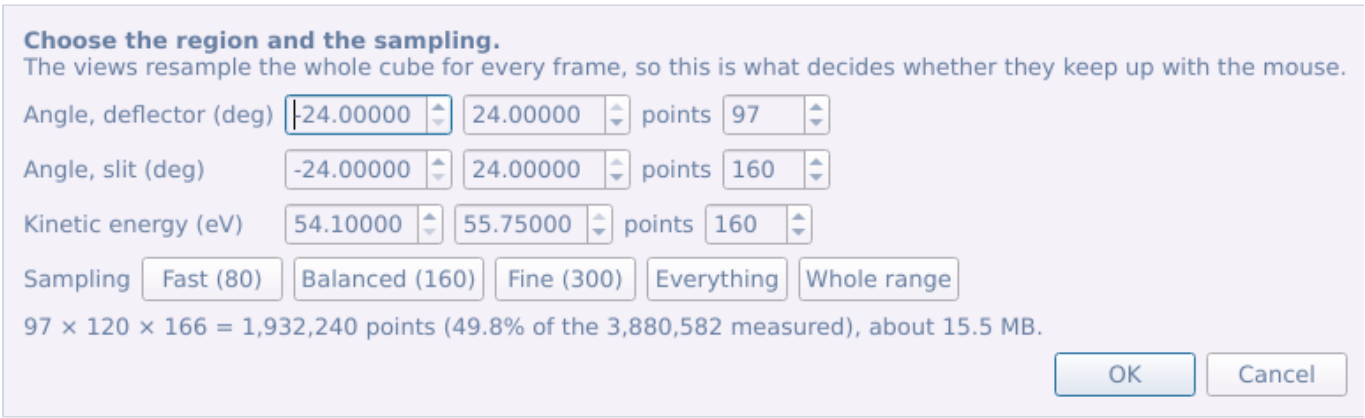
Channels can be hidden, offset into a waterfall (**Offset**), shown with **Error bars** (from a σ channel), and **Range** marks an x window used by the operations. *Functions → Data operations* gives **Crop**, **Bin**, **Normalise**, **Subtract a background** (constant, linear, Shirley), **Shift the axis** and **Add counting errors (√N)**; each makes a new dataset and propagates σ.

Spin EDCs. A spin-resolved EDC opens with one curve per spin channel, so the count rates of all channels can be checked as they come in. The polarisation itself is in *Functions → Spin analysis...* (§21).

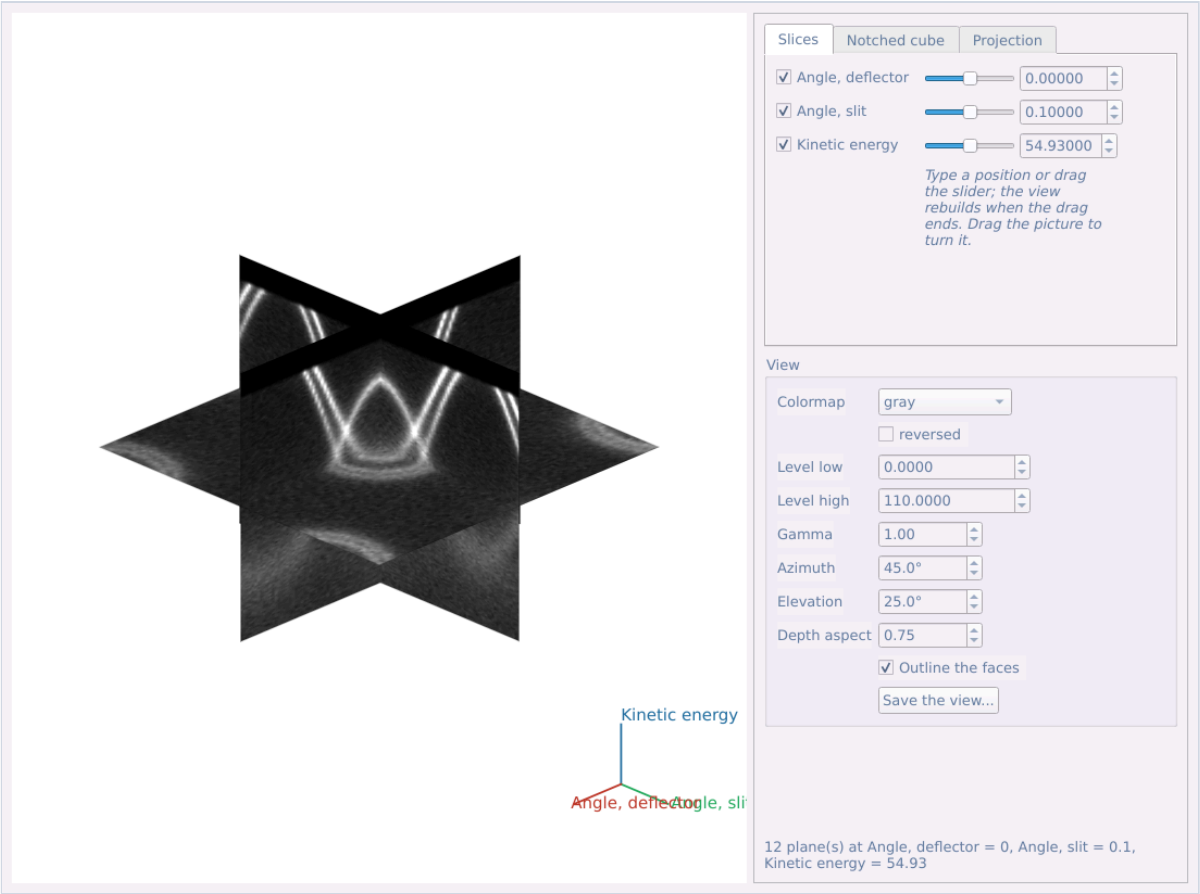


11. The 3-D view of a map

3D view (right-click a map in the list) first asks for the region and the sampling, since every frame resamples the cube — *Fast*, *Balanced* or *Fine* decide whether it keeps up with the mouse.



Then drag the picture to turn it. **Slices** shows three orthogonal planes at the positions you set, **Notched cube** cuts a corner out of the cube, and **Projection** sums along an axis. Colormap, levels, gamma, azimuth, elevation and depth aspect are on the right; **Save the view...** exports the picture.


Slices
Notched cube
Projection
Angle, deflector
0.00000
Angle, slit
0.10000
Kinetic energy
54.93000
Type a position or drag the slider; the view rebuilds when the drag ends. Drag the picture to turn it.
View
Colormap
gray
reversed
Level low
0.0000
Level high
110.0000
Gamma
1.00
Azimuth
45.0°
Elevation
25.0°
Depth aspect
0.75
Outline the faces
Save the view...
Kinetic energy
12 plane(s) at Angle, deflector = 0, Angle, slit = 0.1, Kinetic energy = 54.93

Part III — Advanced processing for publication

12. The usual processing order

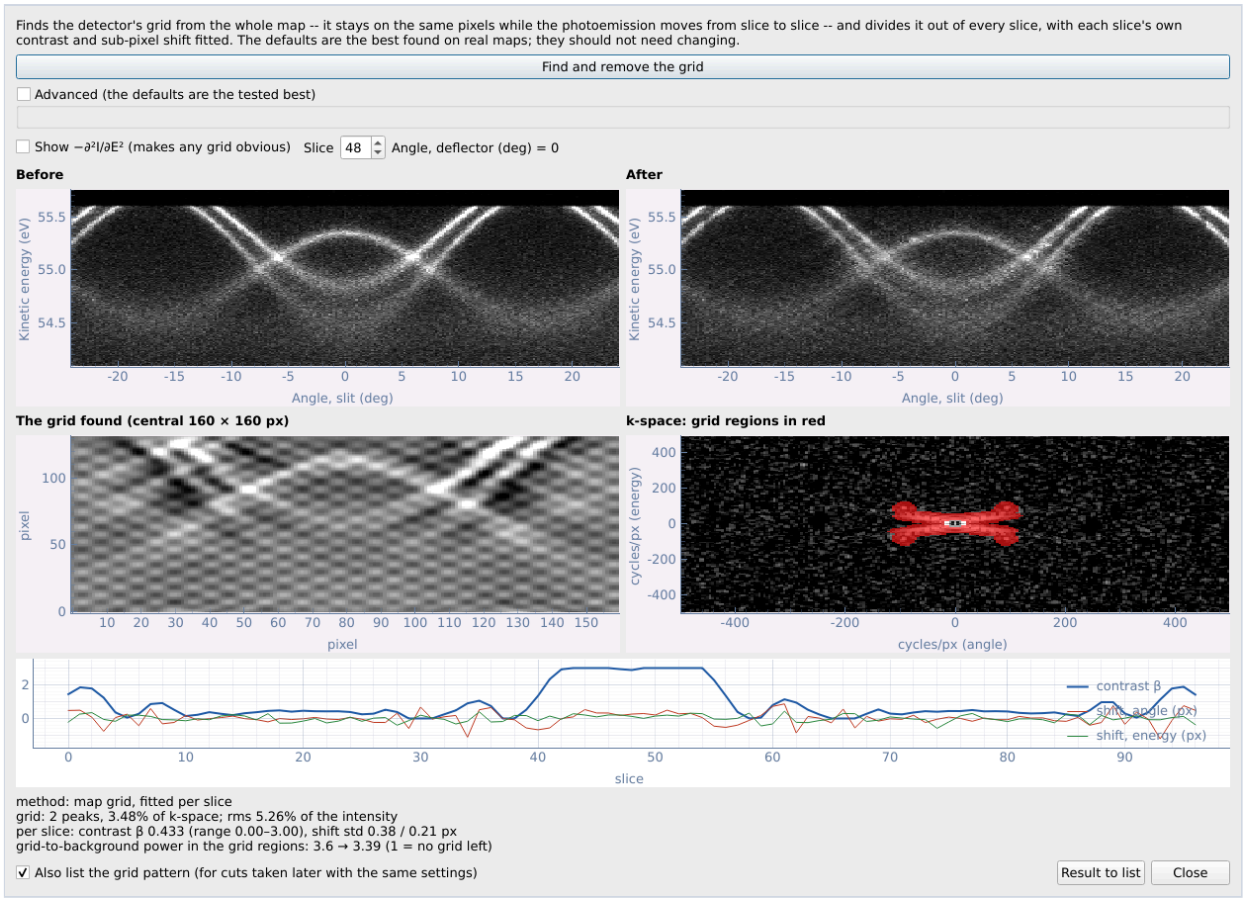
Each step makes a **new dataset** in the list, which records the steps that made it, so the raw data are never changed and every result can be traced. The order matters, because some steps only work on raw detector data:

1. **De-grid** the raw map or cut, if a mesh or MCP pattern shows (§13).
2. **Fermi level**, and **FS correction** if the edge curves across the analyser angle (§14).
3. **k conversion** of the cut or map (§15); for a photon-energy scan, **kz processing** then **kz → momentum** (§17).
4. **Brillouin zone** overlay and **arbitrary cuts** along high-symmetry paths (§16).
5. **Curvature / second derivative** to sharpen bands, **dichroism** (§18, §19).
6. **Peak fits**: dispersion, v_F, m*, self-energy (§20); curve and spin fits (§21).
7. **Figures and export** (§22).

13. Removing the detector grid

MCP (hexagonal) and mesh (square) patterns print on every image. **Do it first**: only data still on the detector's pixels can be de-gridded, so the function refuses anything already converted, corrected or smoothed, and says why.

Functions → De-grid map... on a map (or its slit cut), or *De-grid...* on a cut. For a map, no reference is needed — the grid stays on the same pixels in every slice while the photoemission moves. **Find and remove the grid**:



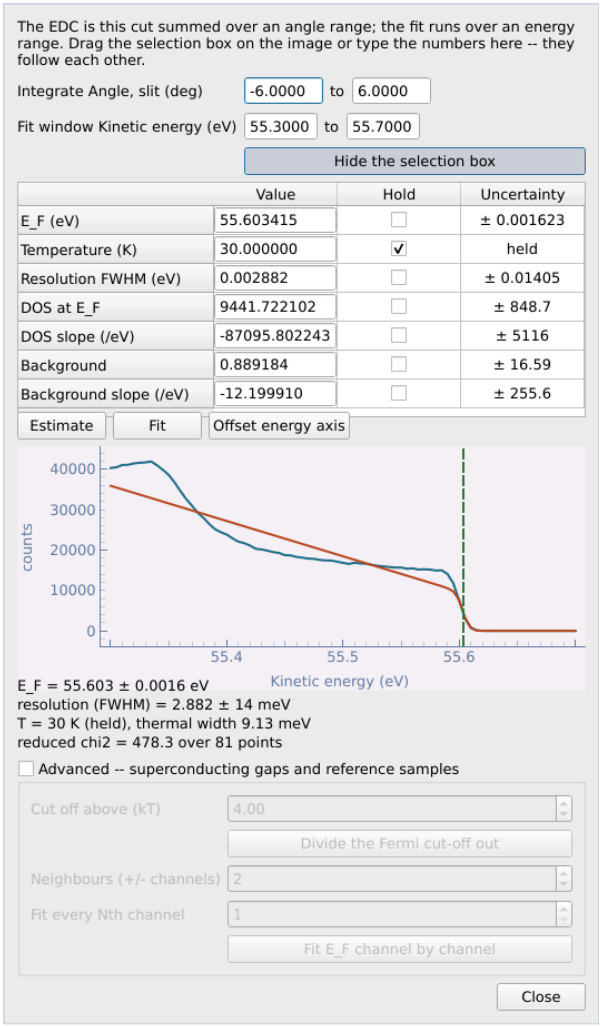
- **Before / After** a chosen slice on one colour scale (tick **Show −∂²I/∂E²** to make any grid obvious).
- **The grid found**, its **k-space regions** (in red), and the **contrast and shift** fitted for every slice.
- The power left in the grid regions (1 = no grid left).
- **Also list the grid pattern** keeps a `[grid]` dataset, which a cut taken later with the same lens mode and pass energy will find and use.

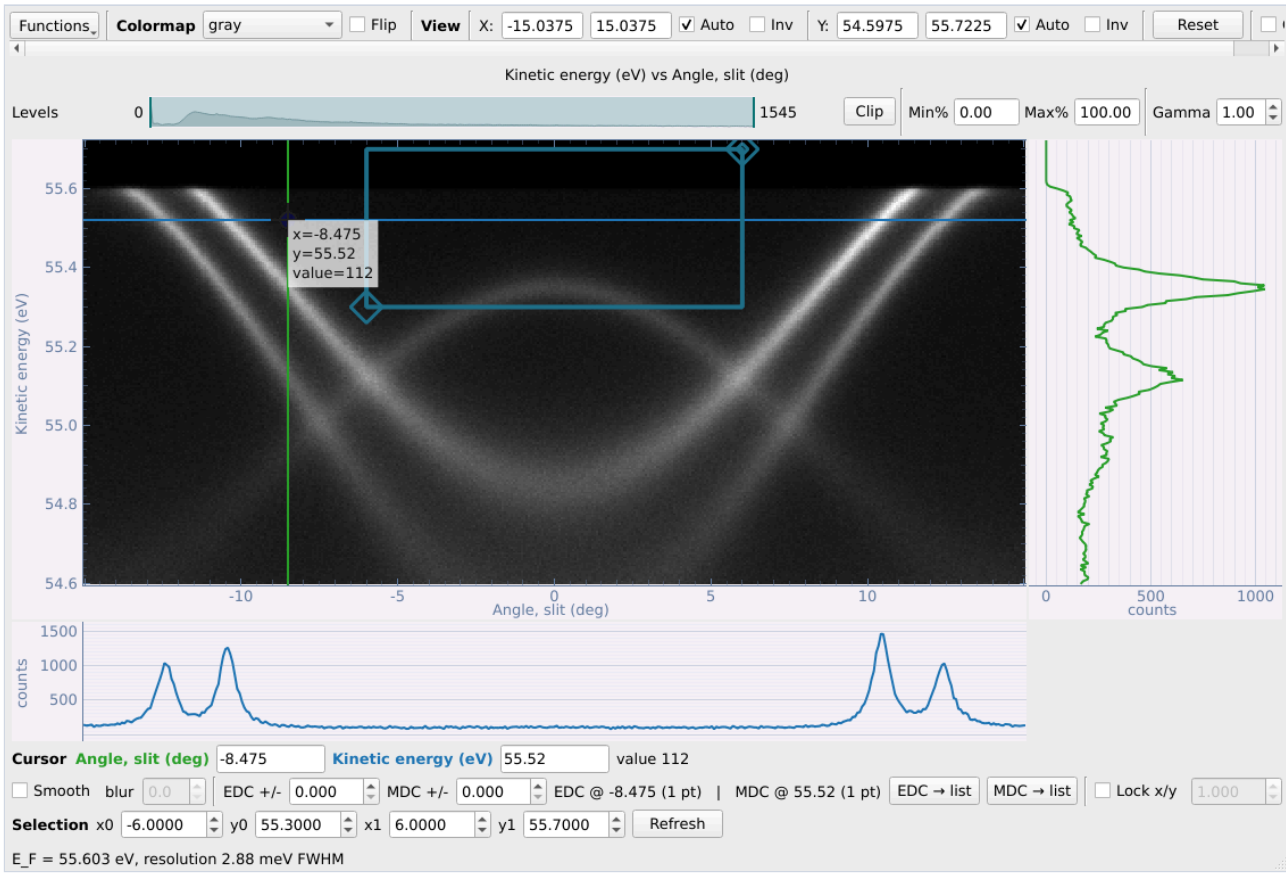
14. Fermi level and Fermi-surface correction

14.1 Fermi level

Functions → Fermi level... fits a Fermi edge (Fermi–Dirac × linear DOS, convolved with a Gaussian resolution) to the EDC summed over a box.

1. Press **Show the selection box** and drag the box on the cut over the edge (or type the angle and energy ranges). The angle range says which channels are summed; the energy range is the fit window.



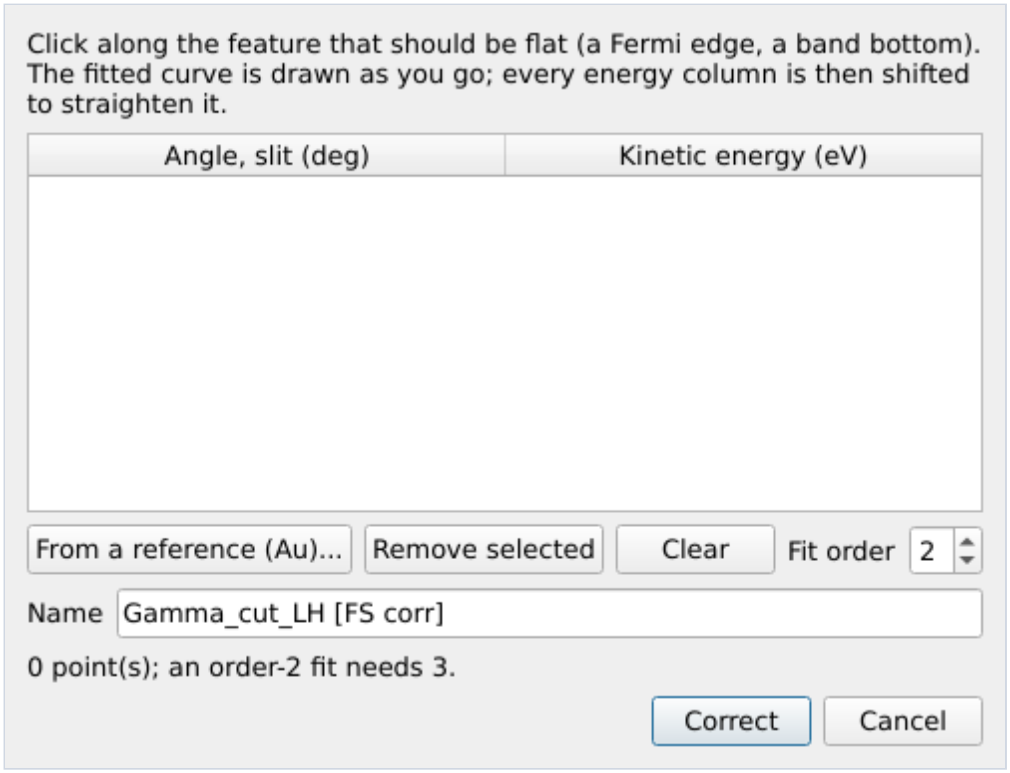


2. **Estimate**, then **Fit**. Tick **Hold** to keep a parameter fixed (the temperature is held at the recorded value by default). Every parameter is reported with its uncertainty.
3. **Offset energy axis** lists a copy of the cut on **E − E_F**.

The *Advanced* corner divides the Fermi cut-off out (for gaps and reference samples) or fits E_F channel by channel.

14.2 Fermi-surface (FS) correction

Functions → FS correction... on a cut (or a map's slit cut) straightens a feature that should be flat — usually the Fermi edge, which curves across the analyser angle. Click along the feature; a polynomial (order 2 by default) is fitted and every angle column is shifted in energy. **From a reference (Au)...** takes the curve from a gold spectrum in the list instead. On a map's slit cut the same shift is applied to the whole cube.

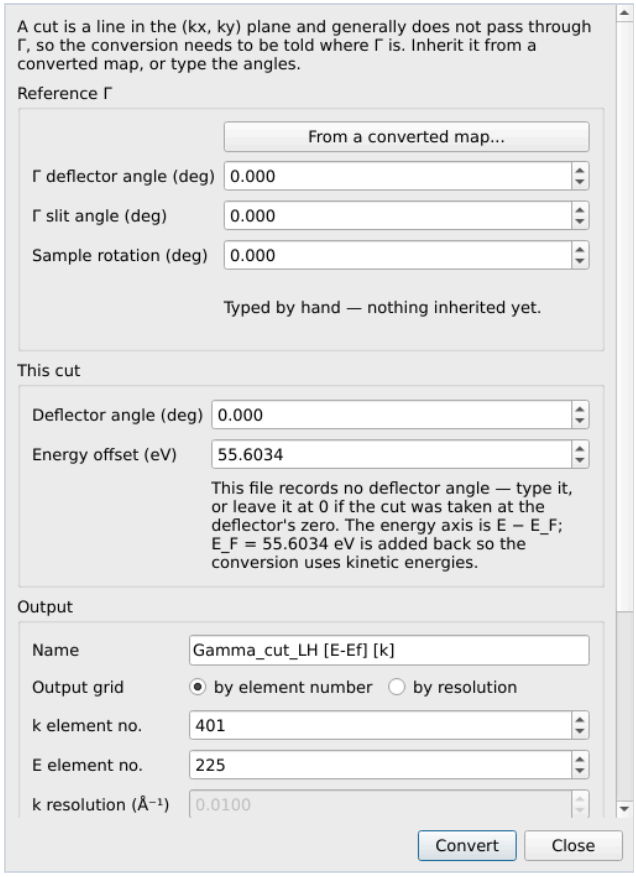


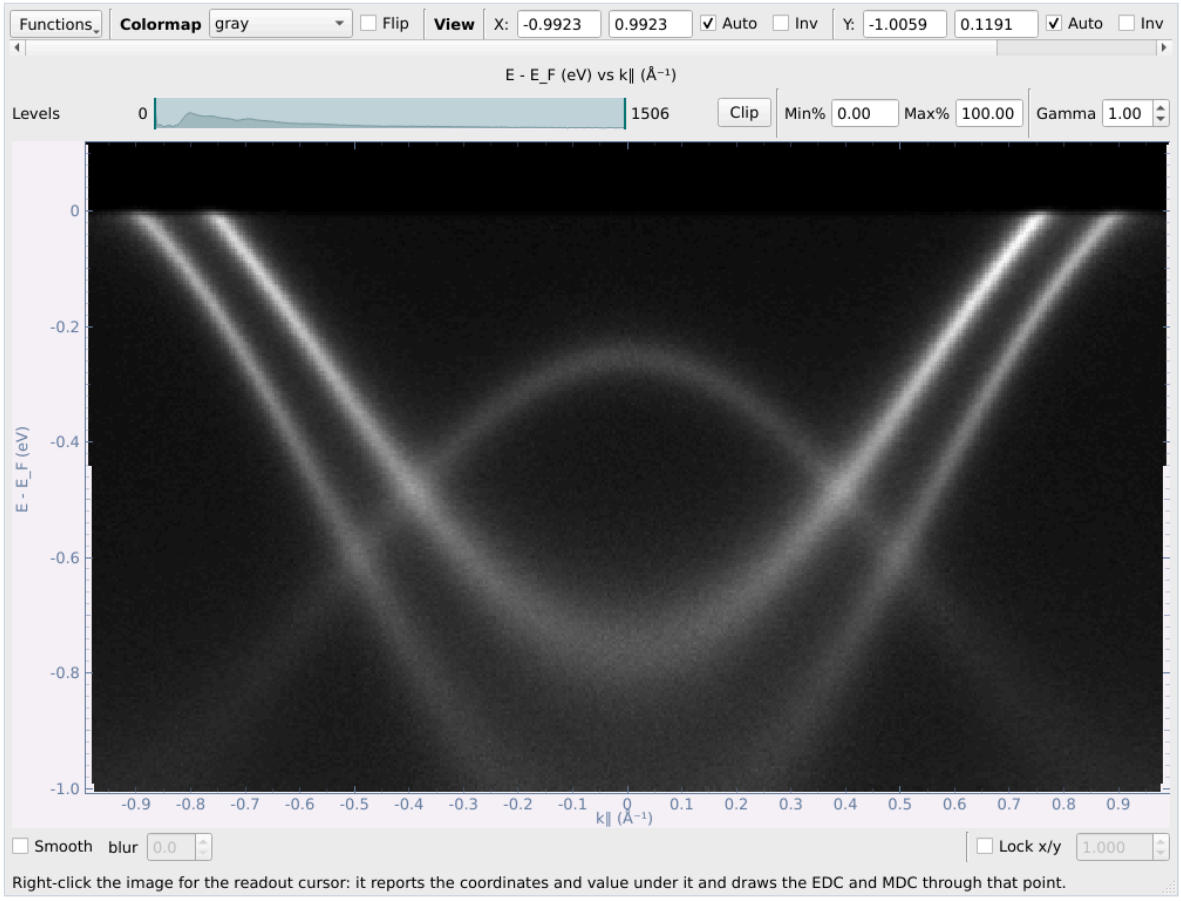
15. From angles to momentum

15.1 Cut k conversion

Functions → Cut k conversion... converts the angle axis to k∥ (Å⁻¹). A cut is a line in the (kx, ky) plane that generally misses Γ, so the dialog asks where Γ is: type the angles, or **From a converted map...** to inherit them from a map already converted.

If the cut is already on E − E_F, the dialog fills in **Energy offset** with the E_F that was subtracted (the note under it says so): the conversion uses true kinetic energies, and the result stays on E − E_F.

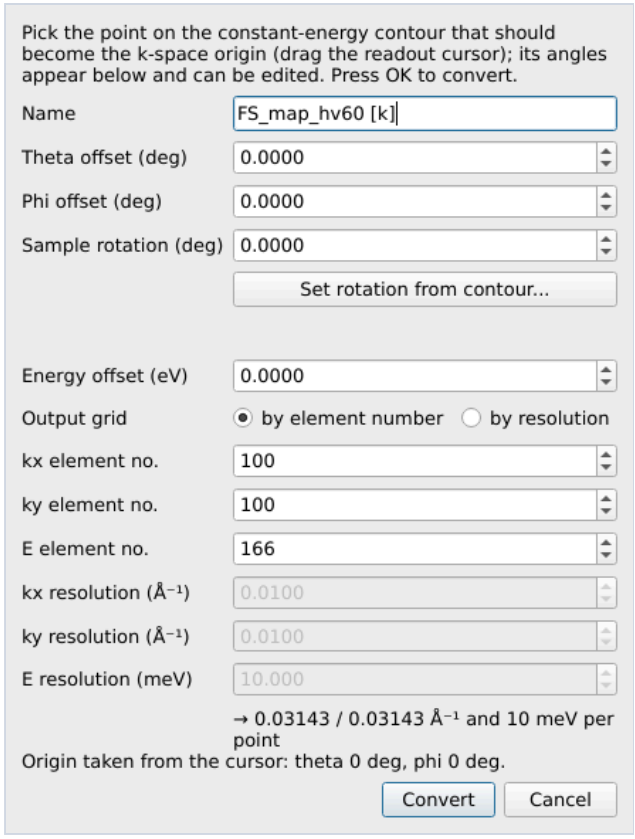


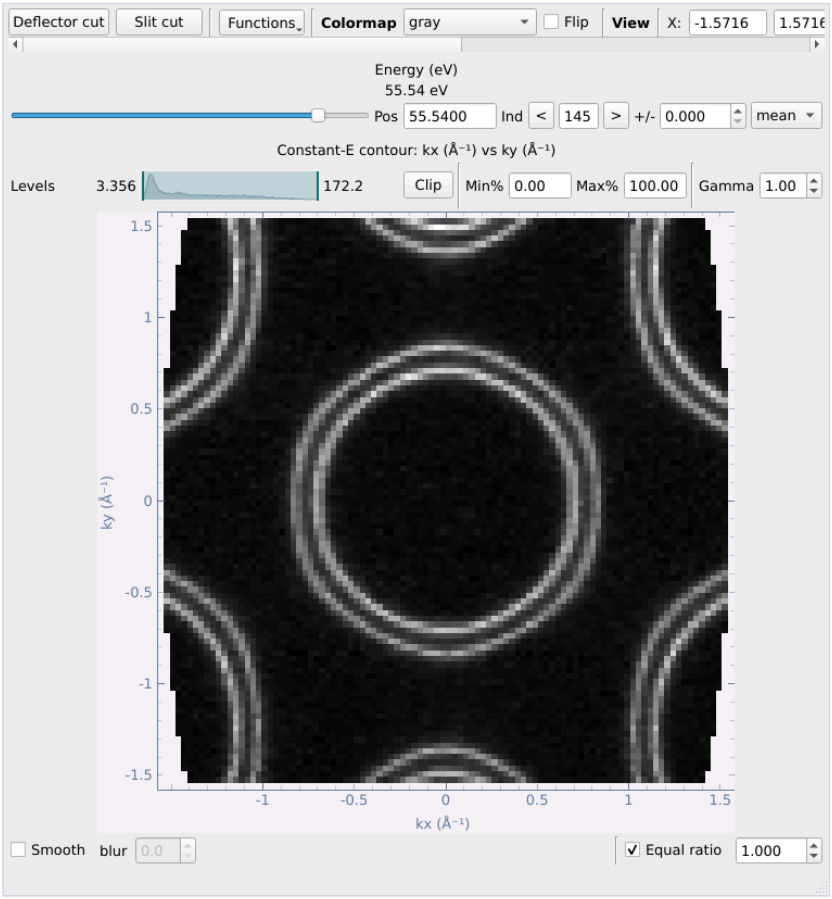


15.2 Map k conversion

Functions → Map k conversion... on the contour window. Drag the readout cursor to the point that should become **k = (0, 0)** — the dialog follows it — then **Convert**. The result is a new `k-map` row in the list.

- **Sample rotation** turns the map about the surface normal; **Set rotation from contour...** reads it off a direction you click on the contour (two points along a mirror line, say).
- **Output grid**: number of points, or resolution (Å⁻¹, meV).
- **Energy offset** is added to the energy axis before converting, for an axis that is not a true kinetic energy (filled in automatically for a map on E − E_F).



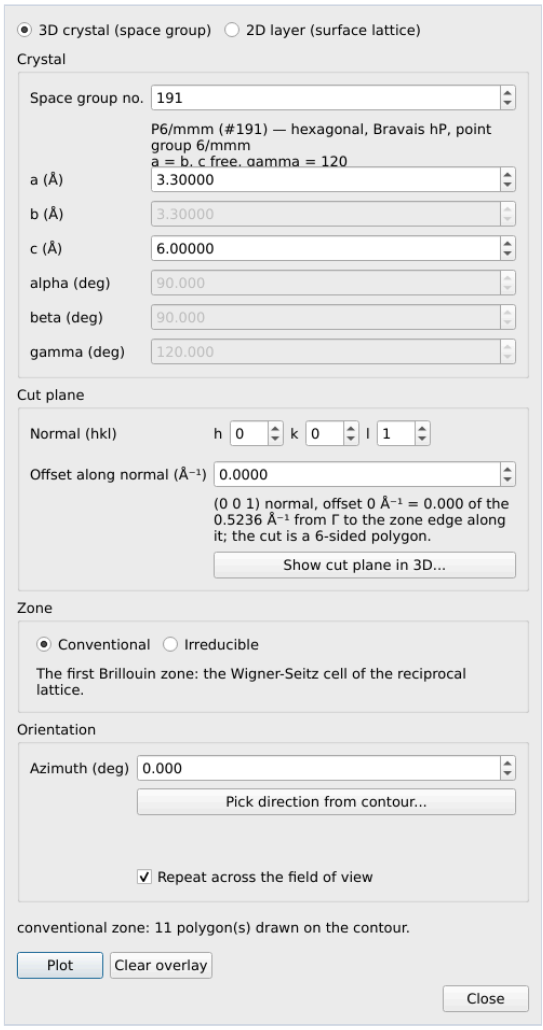


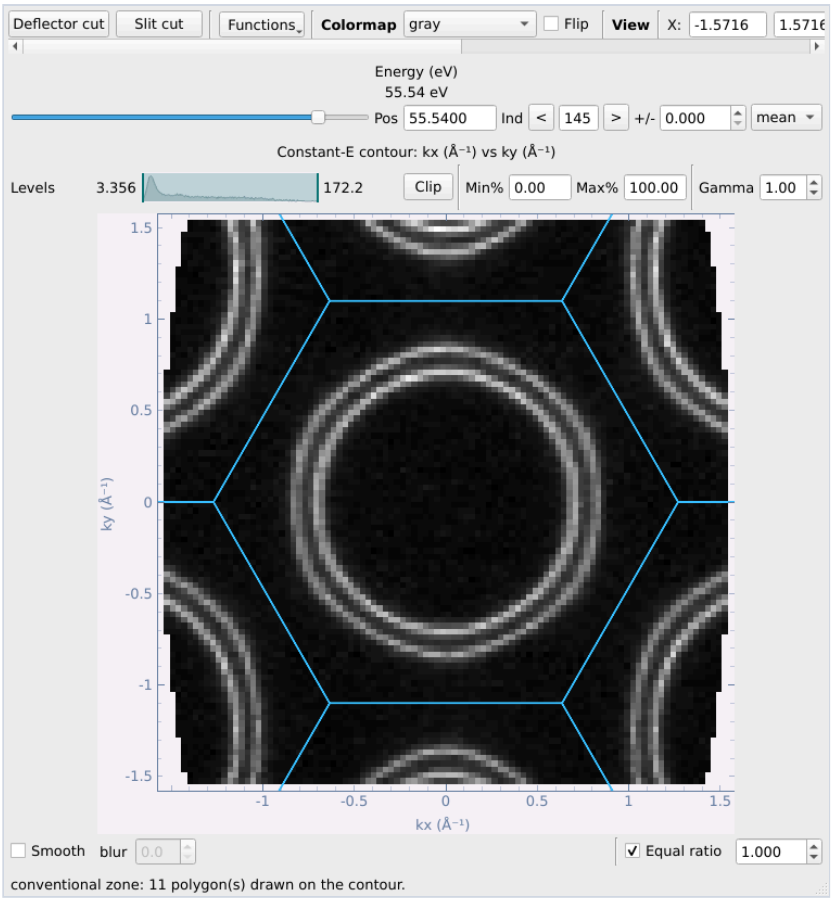
16. The Brillouin zone and arbitrary cuts

16.1 Brillouin zone

On a k-map: *Functions → Visualization → Brillouin zone...*.

1. **Space group** first (1–230); the lattice parameters it fixes are greyed and follow the free ones. Or switch to **2D layer** for a surface lattice.
2. **Cut plane**: the surface normal (hkl) and an offset along it — the cut of the 3-D zone by that plane is drawn, for any surface, not only (001). **Show cut plane in 3D...** previews it (drag to turn it, right-drag to zoom).
3. **Conventional** or **Irreducible** zone; **Azimuth** (or **Pick direction from contour...**); **Repeat across the field of view**.
4. **Moiré** adds a second, twisted or mismatched layer and its moiré zone.
5. **Plot**.

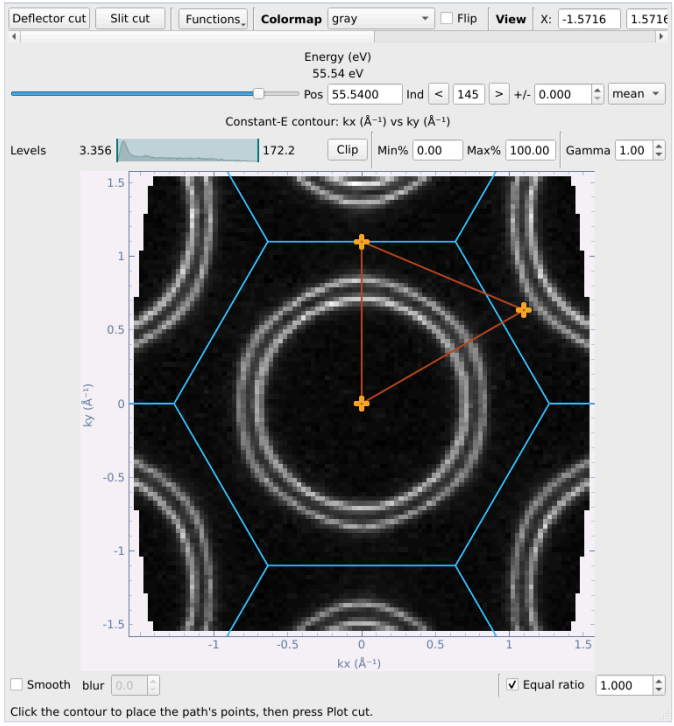




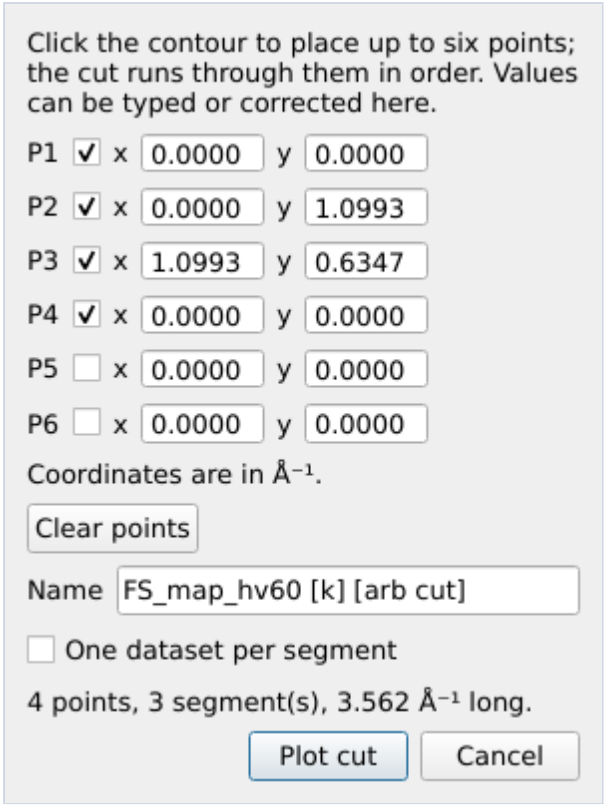
16.2 Arbitrary cut

Functions → Arbitrary cut… on any contour: click up to six points (or type them) and the cut runs through them in order — here Γ → M → K → Γ on the k-map. **One dataset per segment** lists each leg separately.

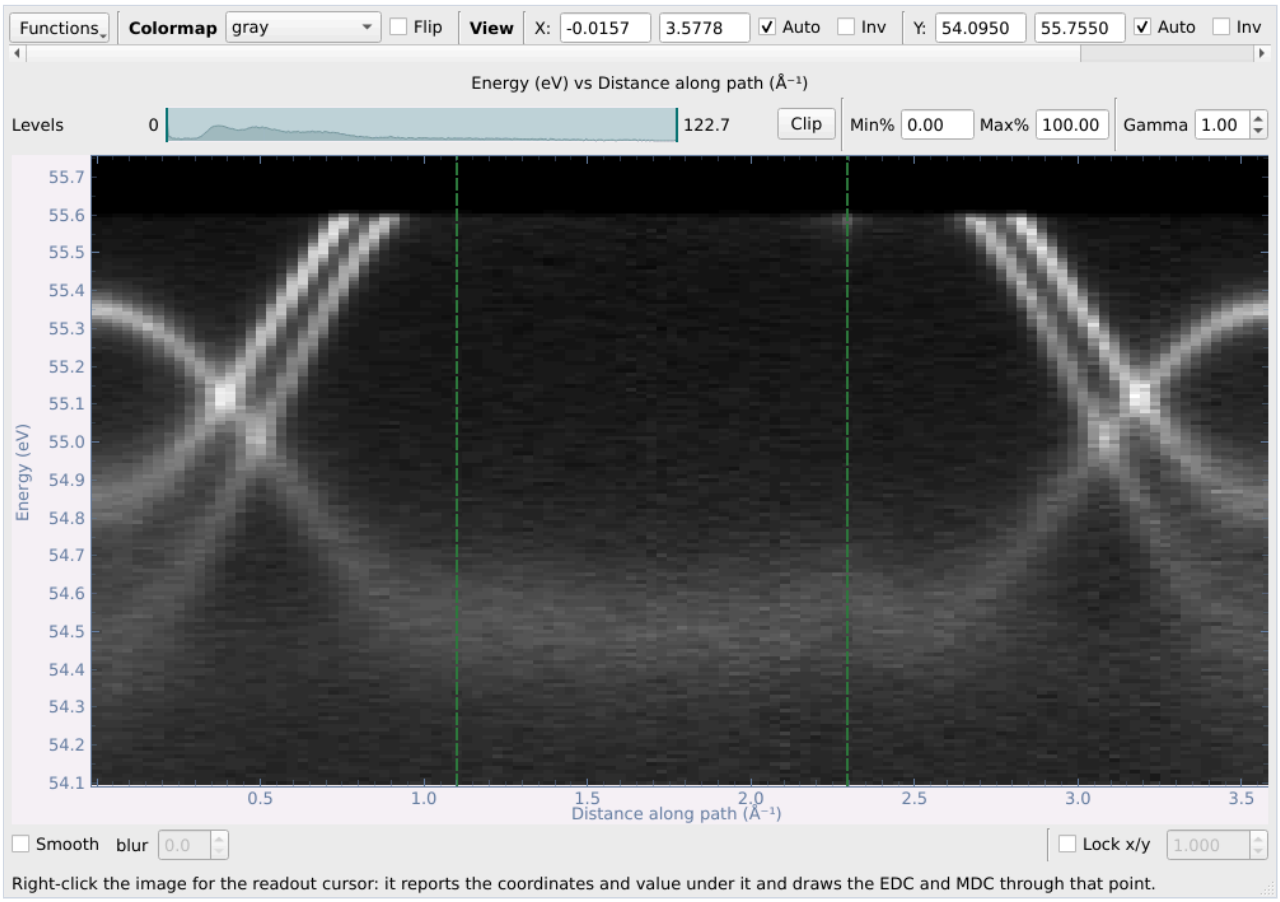
Path on the contour



Points



The result is a cut against the distance along the path; the corners of the path are marked with dashed green lines, so a kink there is not mistaken for physics.

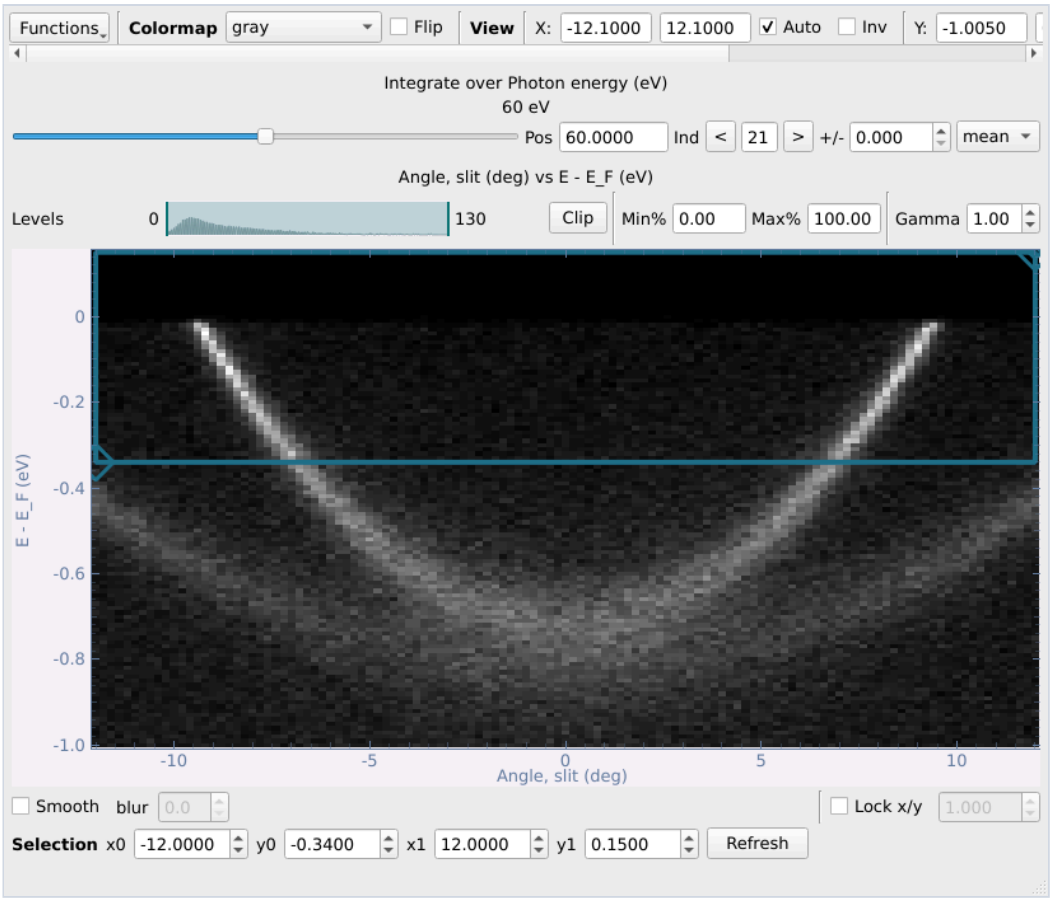


17. Photon-energy scans: kz processing and kz → momentum

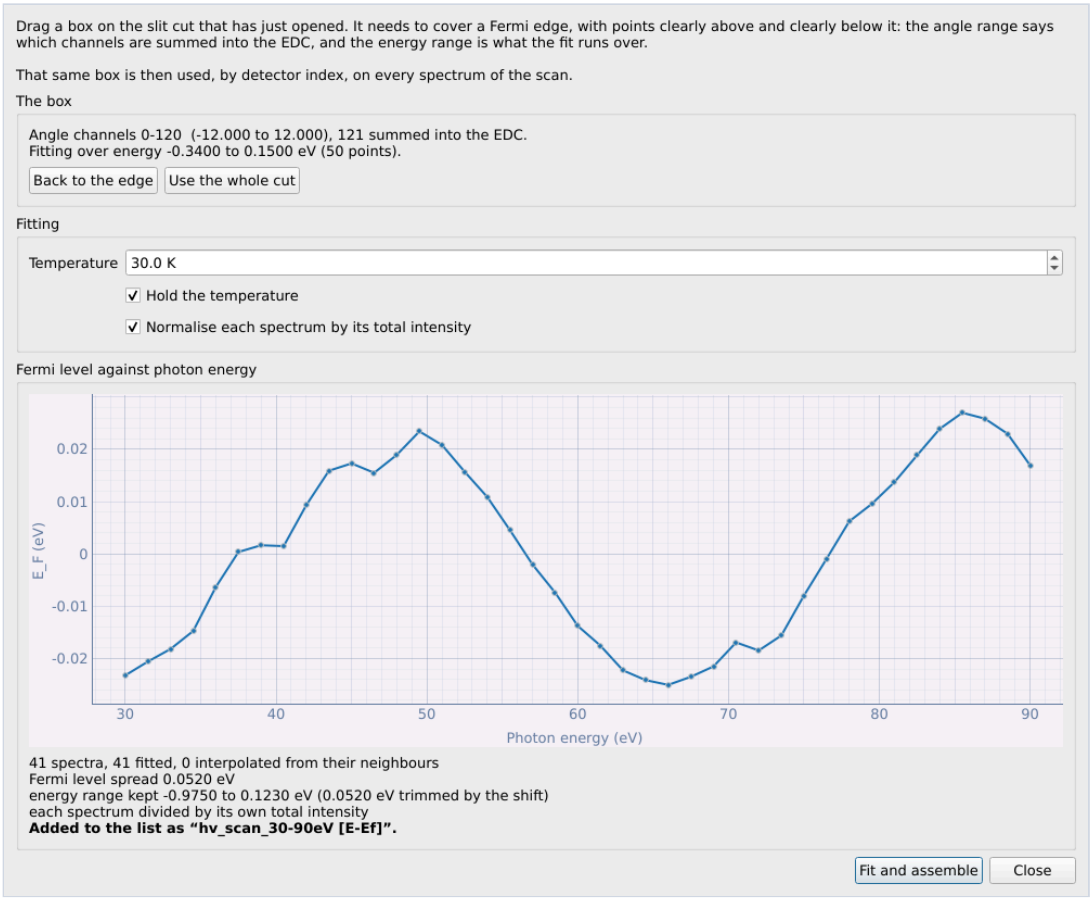
17.1 kz map processing

Nothing in the files says where each spectrum's Fermi level really is. *Functions → kz map processing...* measures it from the spectra themselves.

1. The slit cut opens with a box on it. Drag the box over a Fermi edge — the angle range sets the channels summed, the energy range the fit window.



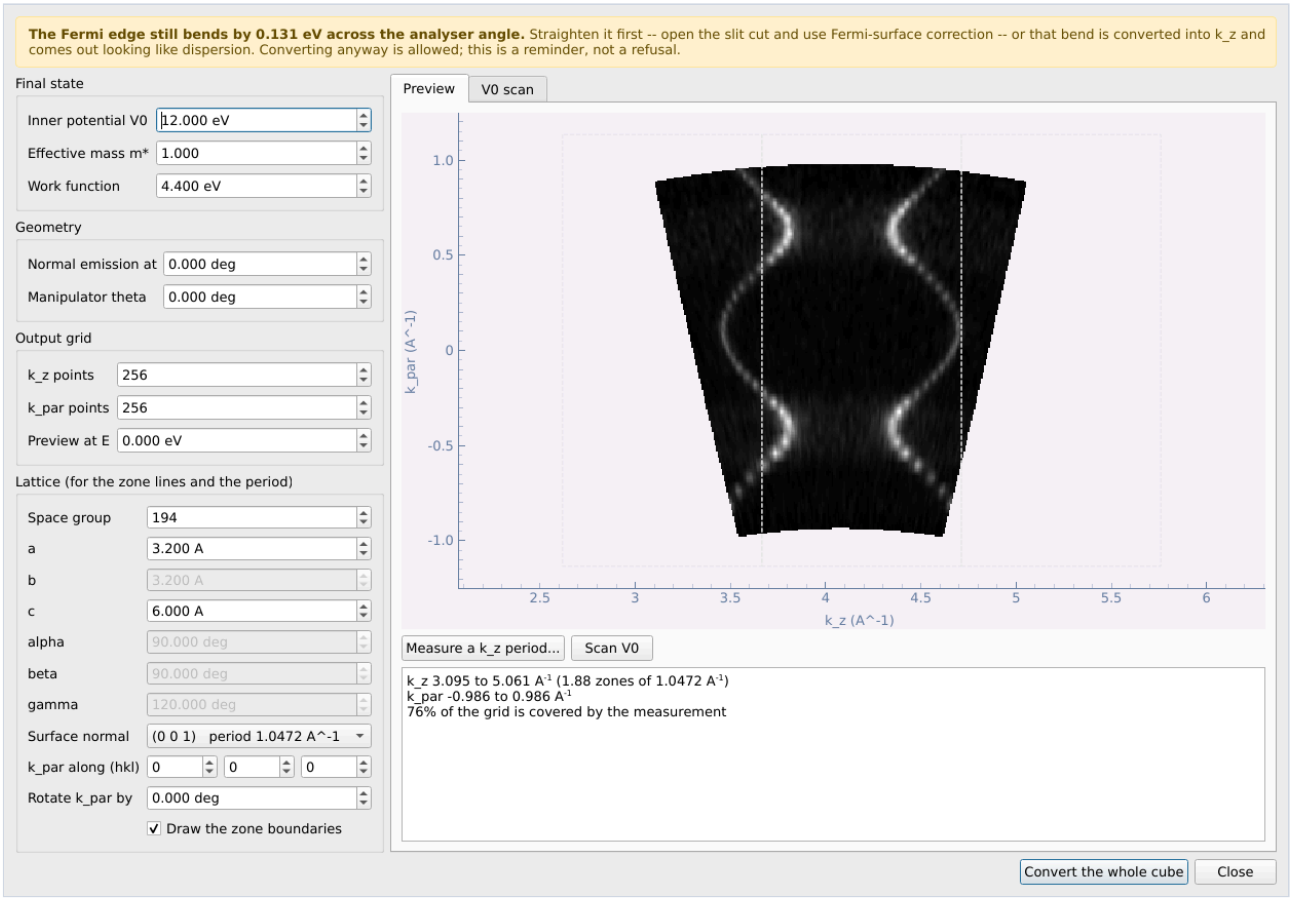
2. **Fit and assemble**: the same box (by detector index) is used for every spectrum; each is shifted to put its own edge at zero; the cube is cropped to the range they all cover; and, by default, each spectrum is **normalised by its total intensity**, which divides out the photon flux.



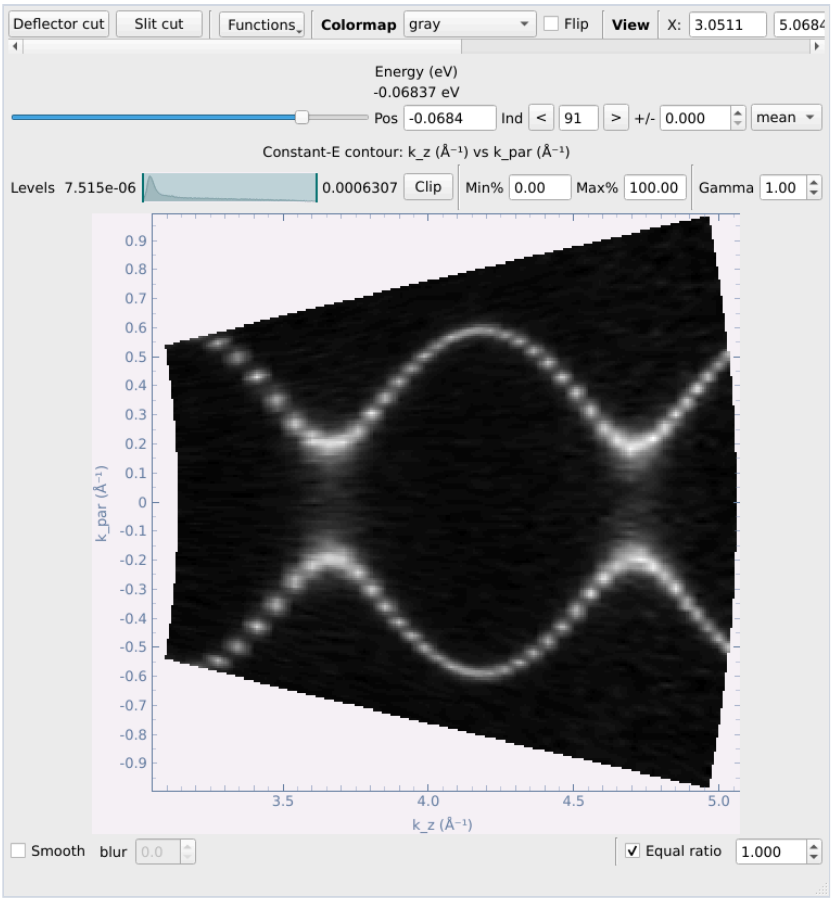
E_F against photon energy is plotted. A smooth curve means the fits are sound; points whose fit is not believed are marked and filled in from their neighbours.

17.2 kz → momentum

Functions → kz -> momentum... converts the scan to (k_z, k∥).

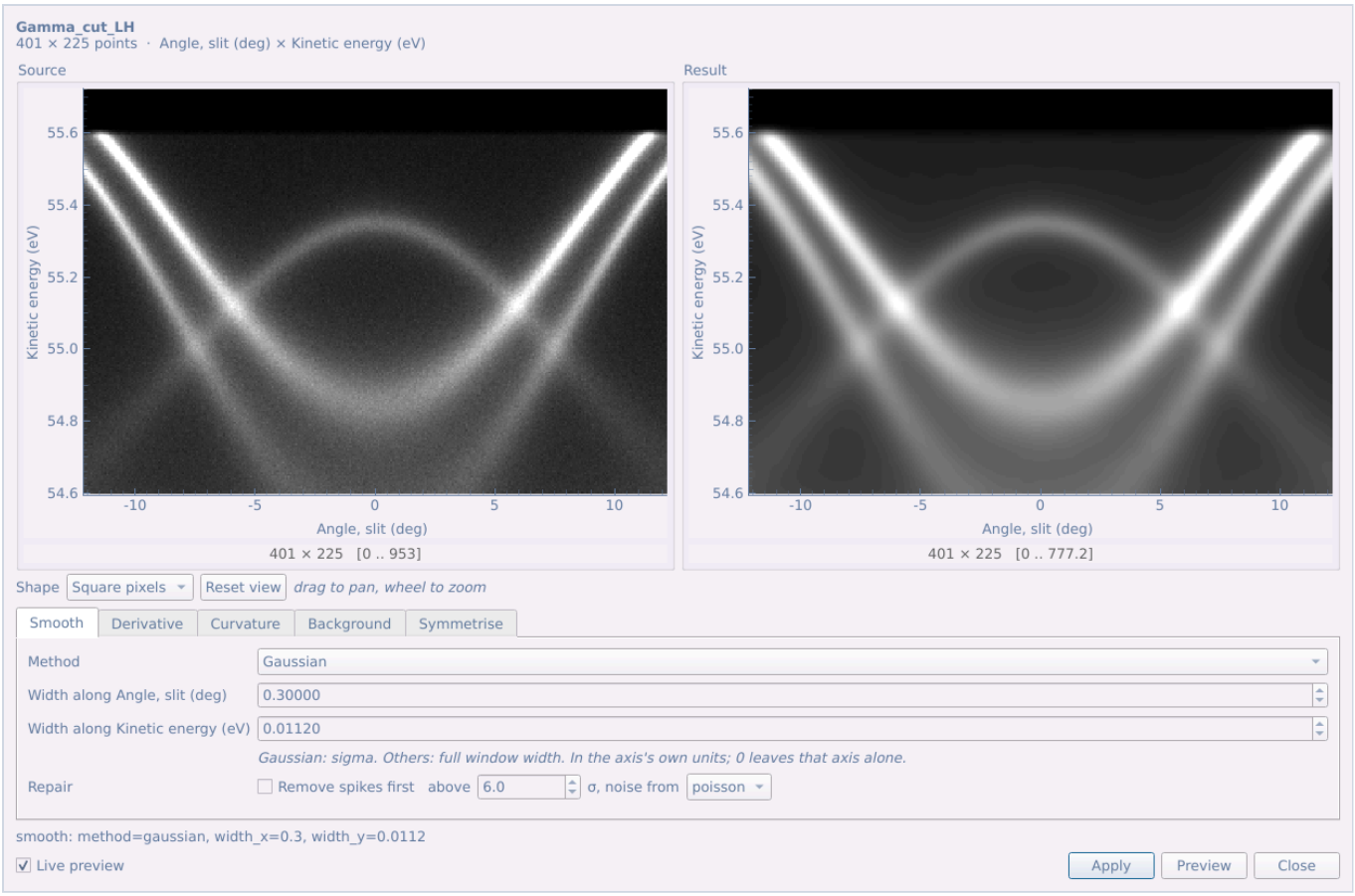


- **Inner potential V0** is the unknown: the preview reconverts one energy slice as you change it, in milliseconds.
- With the **lattice** (space group, a, b, c), the **surface normal** and the in-plane direction, the zone boundaries are drawn over the preview.
- **Scan V0** converts at a range of V0 values and reports where the k_z period matches the lattice — and how well that pins V0.
- **Measure a k_z period...**: click two points that are the same feature one zone apart; the matching reciprocal-lattice planes are listed.
- The yellow banner warns when the Fermi edge still bends across the analyser angle — correct it first (FS correction on the slit cut).
- **Convert the whole cube** lists a `kz map (k)`.



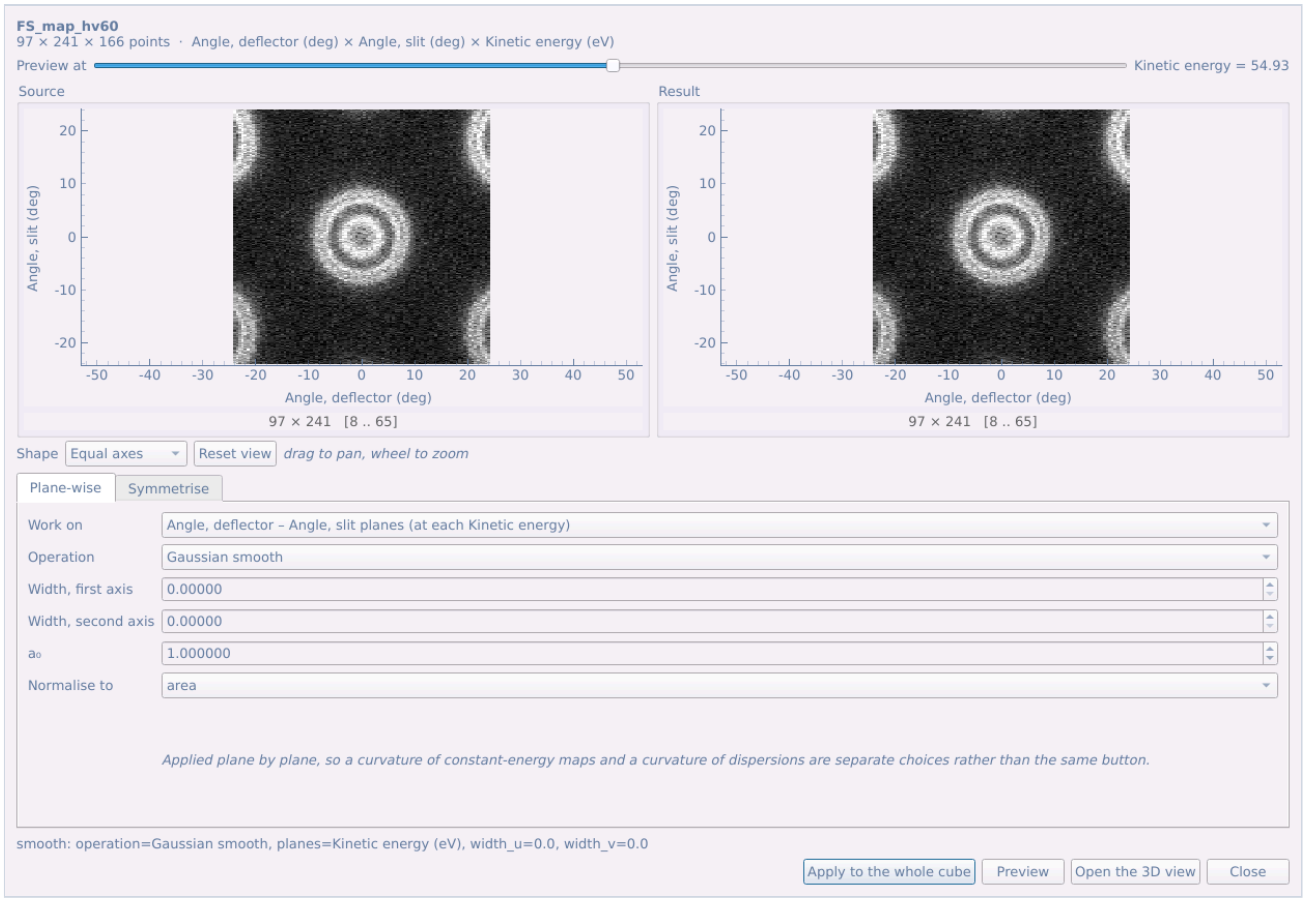
18. Processing: smoothing, derivatives, curvature

Select one or more cuts (or maps) in the list and press **Process…**. Every result is a new dataset carrying the steps that made it.

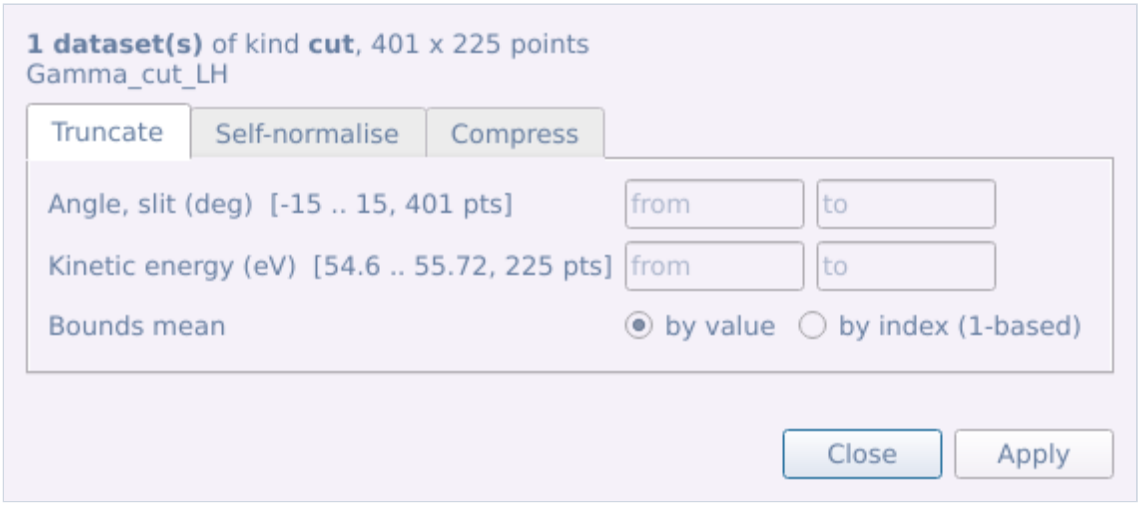


- Tabs: **Smooth** (Gaussian, box, …; optional despiking), **Derivative**, **Curvature** (1-D and 2-D), **Background**, **Symmetrise**.
- **Source** and **Result** are shown side by side on the same scale, so you can see when a setting has eaten the band. They pan and zoom together.
- **Shape**: *Square pixels* (one data point is square — undistorted for an angle-vs-energy cut), *Equal axes* (one unit is the same length on both axes — right for angle-vs-angle or k-vs-k; chosen automatically when both axes have the same unit), or *Fill the box*.
- **Live preview** re-computes as you change a value; **Apply** lists the result.

For a map the panel works plane by plane, and asks **which planes** first — constant-energy maps or dispersions — since the curvature of a contour and the curvature of a cut are different quantities. **Preview at** picks the plane shown; **Apply to the whole cube** runs it on every plane.

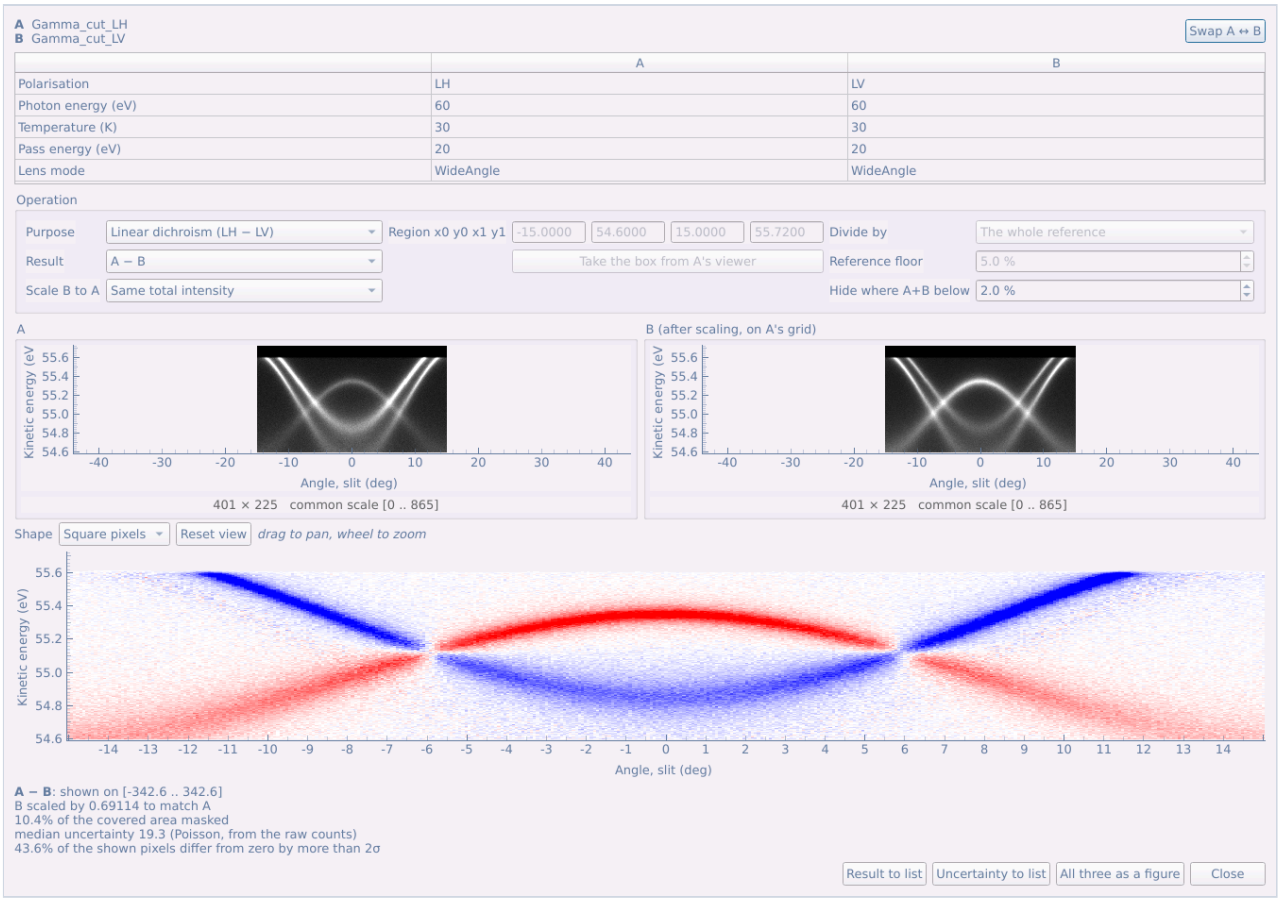


Data operations... (main window) truncates, self-normalises or compresses (bins) one or several datasets at once — as long as they have the same kind and the same number of points on every axis.



19. Cut arithmetic: dichroism and reference division

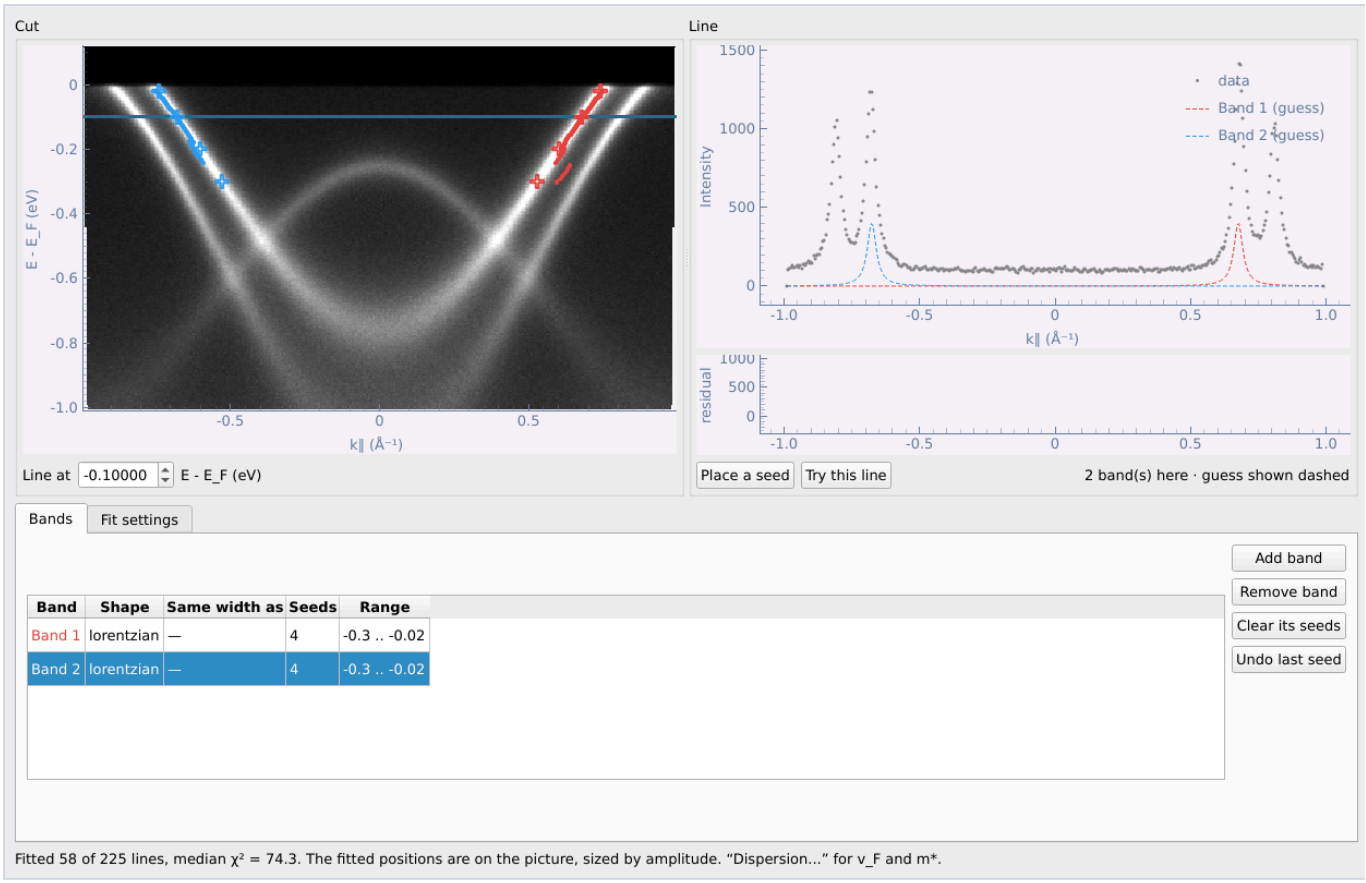
Two cuts combined into a third: linear dichroism (LH − LV), circular dichroism, dividing by a gold reference, or plain differences, ratios and sums. Select two cuts in the list → right-click → **Cut arithmetic on the two...** (the first selected is A), or in a cut's viewer *Functions → Cut arithmetic...* and then click the other cut in the list.



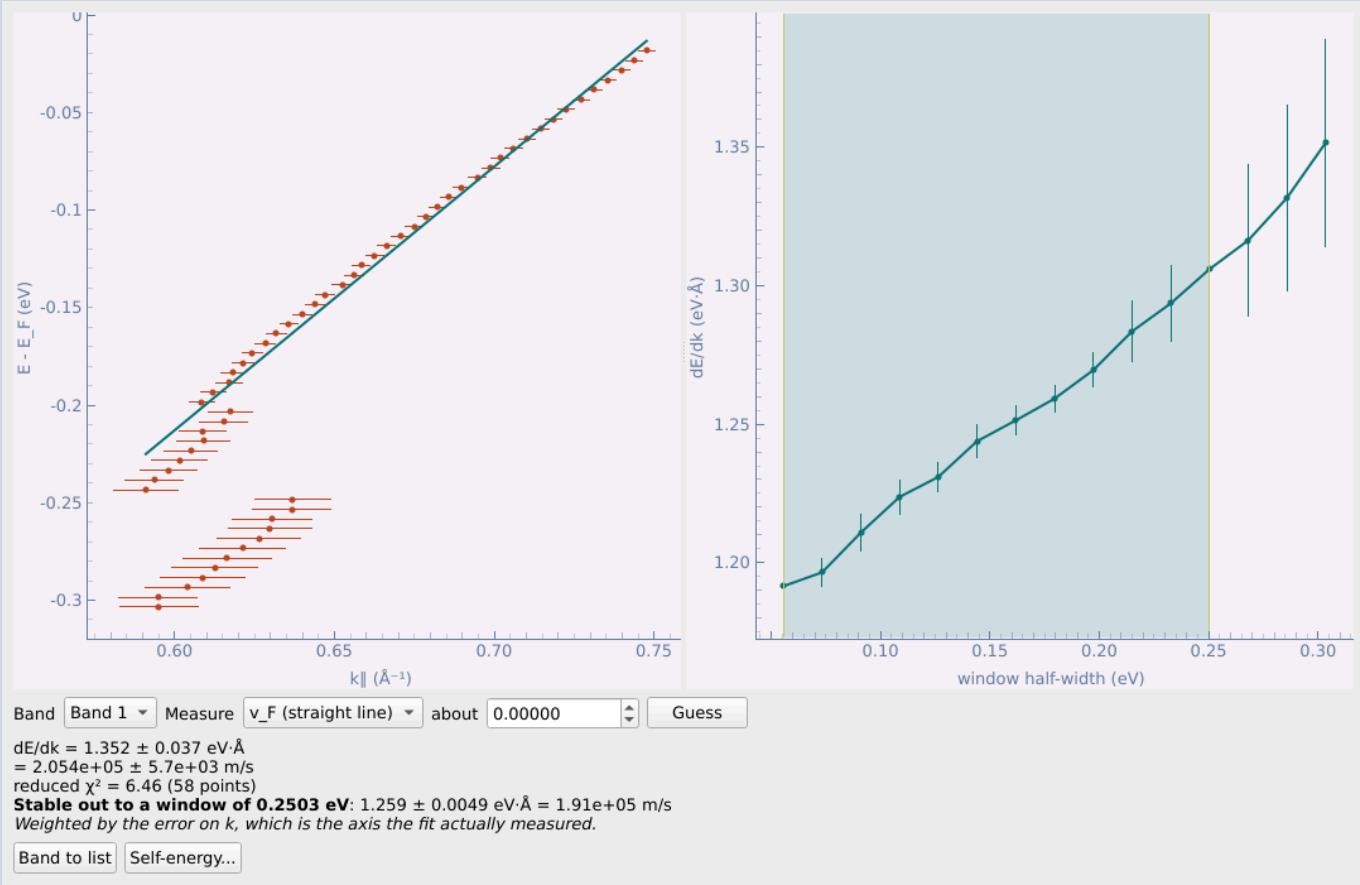
- **The table at the top** compares what the two measurements recorded (polarisation, photon energy, angles, sample position, temperature, pass energy, lens mode). Rows in orange differ beyond a tolerance — a dichroism map between two spots on the sample is a map of *that* difference.
- **Purpose** presets: *Linear dichroism* (A − B, B scaled to the same total intensity), *Circular dichroism* ((A − B)/(A + B)), *Divide by a reference*, *Custom*. Warnings appear if the polarisations look swapped (**Swap A ↔ B**).
- **Scale B to A** by total intensity or by a region; **Hide where A+B below** blanks the background.
- A and B are drawn on one colour scale; the result on a red-white-blue scale symmetric about zero. With raw counts on both, the Poisson uncertainty is propagated and summarised under the plot.
- **Result to list**, **Uncertainty to list**, **All three as a figure**.

20. MDC / EDC fit: dispersion, v_F, m*, self-energy

Functions → MDC / EDC fit... (or right-click the row → **MDC / EDC fit...**) fits peaks line by line and reads the band off the fitted centres. It only accepts a cut whose axis is already in $Å^{-1}$ — a velocity in eV·deg would be meaningless — so convert the cut first (§15).



1. **Bands** tab: **Add band** for each branch; select a band, press **Place a seed** and click the peak on the curve (right). Four or five seeds down a band are enough; **Line at** moves the line being looked at.
2. **Try this line** shows the fit of the current line (dashed: the guess).
3. **Fit settings** tab: MDC or EDC, peak shape (Lorentzian, Gaussian, Voigt), background, step, then **Fit the series**. Fitted positions are drawn on the image, sized by amplitude.
4. **Dispersion...** fits the band: **v_F** (straight line), **m*** (parabola) — and checks how far out the fit is stable. **Self-energy...** gives Re Σ and Im Σ.

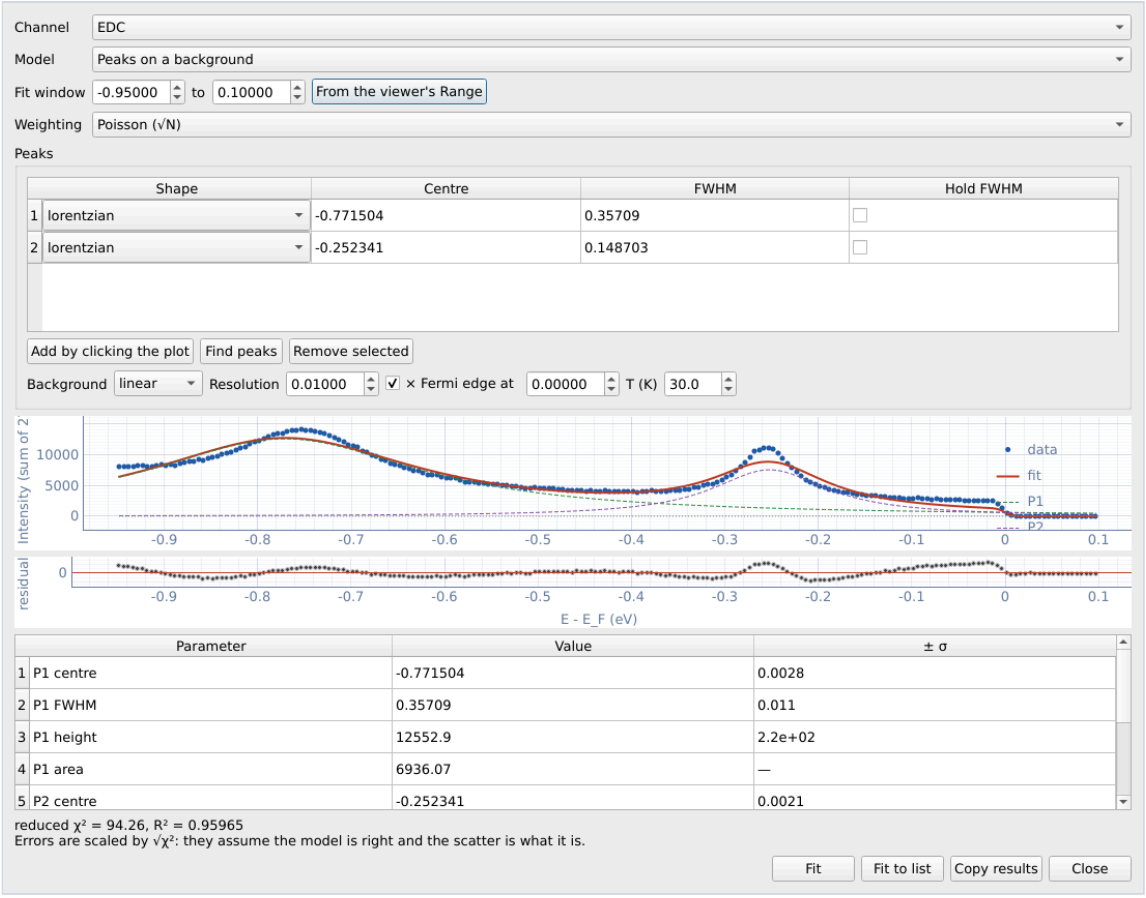


Save table... writes every fitted line; **Band to list** puts a band in the main list.

21. Curve fit and spin analysis

21.1 Curve fit

On a curve: *Functions → Curve fit...*. **Peaks on a background** (Lorentzian, Gaussian or Voigt peaks placed by clicking or **Find peaks**; none / constant / linear / Shirley background; instrumental resolution; optional **× Fermi edge**), or the **Fermi edge** itself. Weighting is the channel's own σ, √N, or uniform. Every parameter comes with its error; the residual is shown in units of σ.



Fit to list keeps data, fit, residual and components; **Copy results** copies a tab-separated table. A Fermi-edge fit can **List a copy with E_F = 0**.

21.2 Spin analysis

On a spin EDC: *Functions → Spin analysis…*:

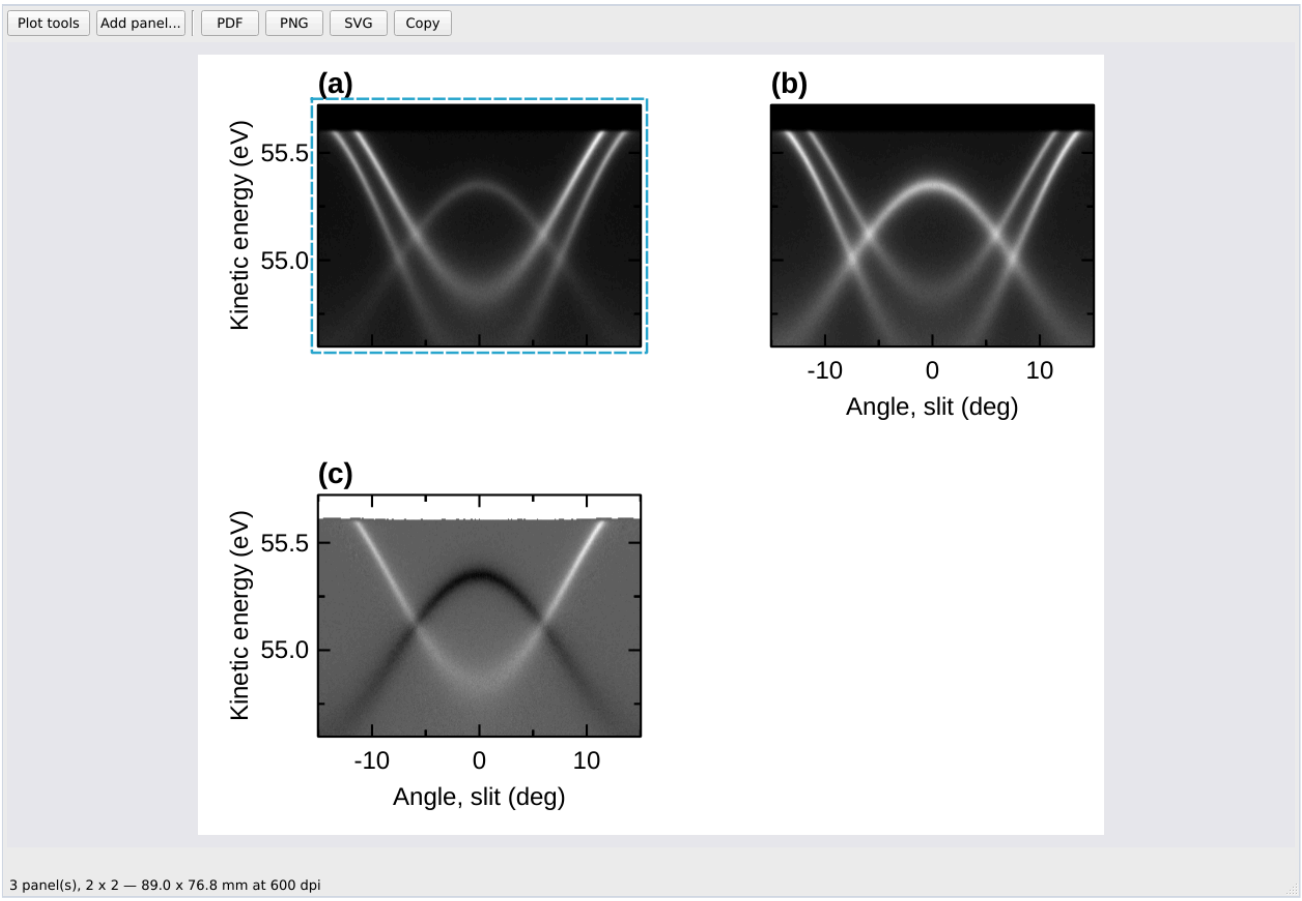


- The table reads each channel's label (manipulator setting, target magnetisation, axis and sign counted as "up"); **Axis** and **Counts as** can be overridden.
- **Cross ratio** (default when the channels allow it) cancels both the transmission and the target reflectivity exactly; **one pair** is available but carries the instrumental asymmetry, which is measured and reported (ε).
- **S_eff** (Sherman function, default 0.2), **Bin**, optional **Zero reference** window.
- Plots: raw channels, P with its error bars, and I↑ / I↓ with errors.
- **P to list**, **I↑ / I↓ to list**, **As a figure**.

22. Figures, export and saving

22.1 Figure composer

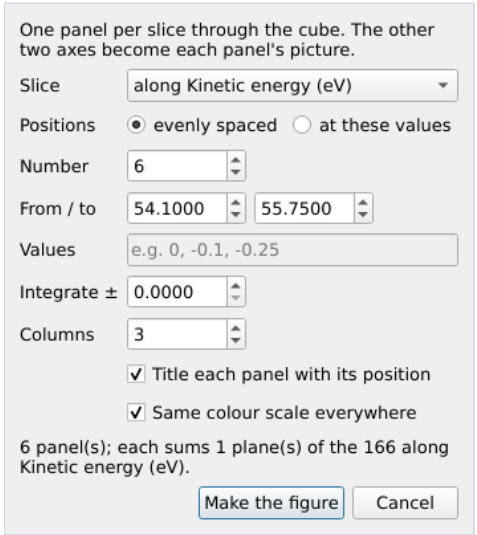
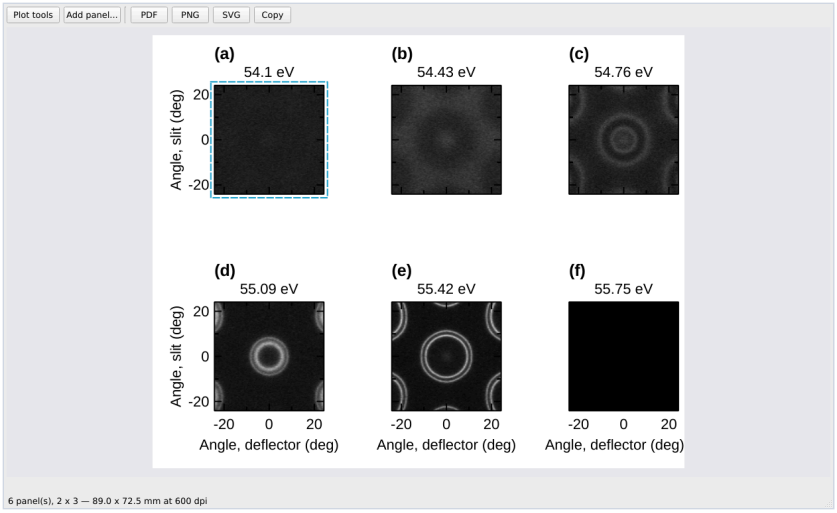
Select cuts (or curves) in the list → right-click → **Plot as a figure**. The composer lays panels out for a paper: one size per column, labels only on the outer edges, (a)/(b)/(c) letters, shared colour scales if you want them.



Plot tools sets everything (size in mm, fonts, ticks, colour bars, lines, arrows, text, insets, scale bars); **Add panel...** adds another dataset; **PDF**, **PNG**, **SVG** export, **Copy** puts it on the clipboard. A stack plot (**Send to figure composer**) and a dichroism result (**All three as a figure**) come here too.

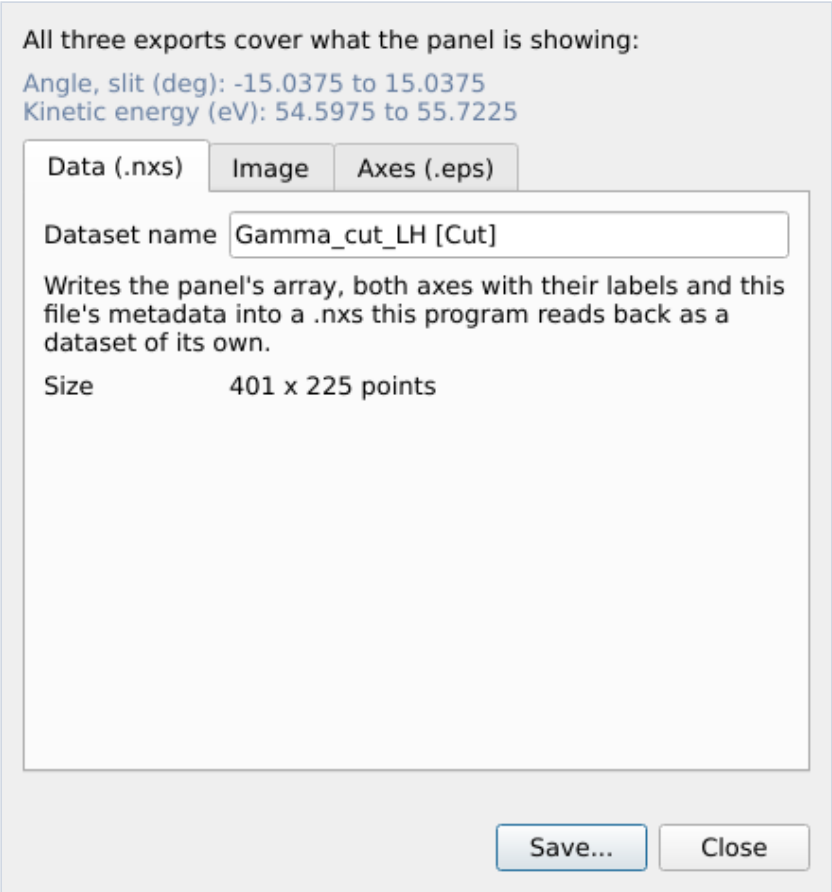
22.2 Slice figure

From a map: *Functions → Slice figure...* lays out a series of slices through the cube — constant-energy contours, or cuts — evenly spaced or at chosen values, with integration, one shared colour scale and titles.

Settings	Result
	

22.3 Exporting a panel

Right-click any image → **Export** → **Data (.nxs)...**, **Image...** or **Axes only (.eps)...**. It always exports exactly what the panel shows (range and all): the data as a dataset this program reads back; an image with the colormap applied (PNG, TIFF or JPEG, at a chosen size and dpi, with or without the axes); or the frame, ticks and labels as vector PostScript to place the image in, in Illustrator. In a SPEM window, right-click the panel you want — the map and the spectrum export separately.



22.4 Saving datasets

Right-click rows in the list → **Save dataset(s)...** writes them to a `.nxs` file of your own; several selected rows go into one file, each under its listed name (use **Rename...** first). The saved file opens again with every step that made each dataset, and the save is written to the session log.

Part IV — Troubleshooting

23. Autosave, crash recovery and memory

- **Autosave.** Every computed dataset (a k-map, a processed cut, a fit result…) is written, uncompressed, to this session's folder in `~/.arpes_viewer/sessions/` the moment it exists. Real-space scans are not (they are the measurement itself, and large).
- **A normal close deletes the autosaved copies.** When you close the main window and confirm (everything you want is saved, or *Close anyway*), the open viewers are closed and the session folder is deleted straight away. A dataset's copy also goes as soon as you save it to a file of your own.
- **Recovery after a crash.** Only if the program did not close normally (a crash, a kill, a power cut) is the session folder kept: the next start offers its datasets back, and it is deleted after three days.
- **Operations log.** `~/.arpes_viewer/sessions/operations.log` (or right-click in the list → **Session log…**), one line per autosave, save and deletion, with paths — how to find what was where.
- **Closing** with unsaved computed data asks first, and names the datasets: **Save to a file…**, **Close anyway** (the autosaved copies are deleted) or **Cancel** (nothing happens).
- **Memory box.** This program's memory, the system's, how many viewers and HDF5 files are open, and how much computed data is held. **Free memory** drops cached copies of computed data (they are re-read from their autosaved file when needed); **Close all viewers** closes every window and then frees memory.
- **Long operations** (reading a big map, converting, de-gridding, assembling a folder) run off the GUI thread with a progress bar and a working Cancel.

24. Common problems

Installation and start-up

- **`ModuleNotFoundError: No module named 'PyQt5'`** (or pyqtgraph, h5py…) — the packages are installed in a different Python from the one running the program. Activate the environment (§2) and check with `python -c "import PyQt5, pyqtgraph, h5py, scipy"`.
- **`qt.qpa.plugin: Could not load the Qt platform plugin "xcb"`** (Linux) — a system Qt library is missing; install it with your package manager (`sudo apt install libxcb-xinerama0 libxkbcommon-x11-0` on Ubuntu), or use the conda installation.
- **Fonts or windows too large or too small** (high-DPI screens) — the program follows the system scaling; set `QT_SCALE_FACTOR=0.9` (or another value) in the environment before starting it.
- **The settings look wrong** — delete `~/.arpes_viewer/config.json`.

Reading data

- **A file is misread** — choose the reader explicitly in the loader, and see *Assumptions worth knowing about* in the README (axis order in SOLEIL files).
- **A k conversion is not offered** — the first axis of the map is not an angle (a photon energy, temperature…). Check *First axis of a map is* in the loader.
- **A CASSIOPEE folder is listed as separate spectra** — the `_i` parameter files next to the spectra are missing or renamed; they say what was stepped.

Right-click menus

- **A list entry is greyed out** — hover over it: the tooltip says what the selection lacks (one row, two cuts, a map, a cut in k…).
- **"Integrate selection" is greyed** — draw a box first (right-click → **Selection box**).
- **"Integrate selection" is missing** — it exists only in SPEM windows with a 2-D map; on a line scan, a cut or a map the box is used for the stack plot and the coordinates.
- **The SPEM map stays dark or shows only part of the sample** — it is showing the intensity of a spectrum box (its title says so). Right-click the map → **Spatial map over the whole spectrum**.
- **The SPEM map looks transposed or mirrored** — **Swap X and Y** (right- click the map) and the **Inv** boxes in the view bar change only the drawing; ANTARES scans open the way the beamline draws them.
- **The line profile's line has gone** — it is removed when its window closes; open it again from the right-click menu.
- **The MDC / EDC fit refuses a cut** — its axis is still an angle; convert it first (§15).

Processing

- **"Not available here" in the Functions menu** — hover over the entry: the tooltip says why.
- **De-grid refuses a dataset** — it has already been converted, corrected or smoothed; de-grid the raw data first (§13).
- **Cut k conversion of an E − E_F cut** — the energy offset is filled in for you from the Fermi fit; if you shifted the axis some other way, type the kinetic energy of E_F (photon energy − work function) into *Energy offset*.

Speed and memory

- **Dragging the cursor on a big map is sluggish** — a map read lazily from its file re-slices in about 0.1 s per window, so the other windows wait for the mouse to pause. Once the map has been read into memory it follows live.
- **The machine is paging** — watch the Memory bar (amber past a third of the RAM, red past a half); use **Free memory** or **Close all viewers**.
- **After a crash** — restart the program: it offers the autosaved datasets back. The log (**Session log…**) says where each one is.